

1. Introduction

Most snippets assume the following macros.

```
1 #include <bits/stdc++.h>
2
3 using namespace std;
4
5 #define P push_back
6 #define I insert
7 #define X extract
8 #define L lower_bound
9 #define U upper_bound
10 #define C count
11
12 auto ckmn = [](auto &x, auto y) { return y < x ? x : y, true : false; };
13 auto ckmx = [](auto &x, auto y) { return y > x ? x : y, true : false; };
14 auto lso = [](integral auto n) { return n & (-n); };
15 auto pop = [](integral auto n) { return popcount((unsigned long long)n); };
16
17 #define W(x, y) ((x & y) == y)
18 #define N(i, n) "ln"[i == n]
19 #define Z(x, i) (x >> i & 1)
20 #define F(i, l, r) for (int i = l; i < r; i++)
21 #define G(i, l, r) for (int i = l; i <= r; i++)
22 #define H(i, r, l) for (int i = r; i >= l; i--)
23 #define A(a) (a).begin(), (a).end()
24 #define R(a) (a).rbegin(), (a).rend()
25 #define S(a) (int)ssize(a)
26 #define B(a) (a).begin()
27 #define E(a) (a).end()
28 #define T(a) prev(E(a))
29 #define O(a) sort(A(a)), a.erase(unique(A(a)), E(a))
30
31 template <typename T = int>
32 using V = vector<T>;
33 using ll = long long;
34 using ul = unsigned long long;
35 using ii = array<int, 2>;
36
37 constexpr int oo = 0x3f3f3f3f;
38 constexpr ll OO = 0x3f3f3f3f3f3f3f;
```

1.1. Template (Proton)

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 typedef long long ll;
4 typedef vector<ll> vll;
5 typedef vector<vll> vvll;
6 typedef pair<ll, ll> pll;
7 typedef vector<pll> vpll;
8 typedef vector<int> vi;
9 typedef vector<vi> vvi;
10 typedef pair<int, int> pi;
11 typedef vector<pi> vpi;
12 typedef set<ll> sll;
13 typedef set<int> si;
14 typedef map<ll, ll> mll;
15 typedef map<int, int> mi;
16 typedef vector<bool> vb;
17 #define pb push_back
18 #define INF(dt) numeric_limits<dt>::max()
19 #define NINF(dt) numeric_limits<dt>::min()
```

1.2. Template (Baytoro)

```
1 #include <bits/stdc++.h>
2 using namespace std;
3 #define ll long long
4 #define pb push_back
5 #define fr first
6 #define sc second
7 #define all(x) (x).begin(), (x).end()
8 #define rall(x) (x).rbegin(), (x).rend()
9 #define int ll
```

1.3. I/O

For large inputs, it may be necessary to enable fast I/O.

```
cin.tie(nullptr)->sync_with_stdio(false);
```

Regarding interactive problems, remember to flush the stream using endl!

1.4. Order Statistic Tree

```
1 #include <bits/extc++.h>
2
3 using namespace std;
4
5 using Ost = __gnu_pbds::tree<
6 int, __gnu_pbds::null_type, less<>, __gnu_pbds::rb_tree_tag,
7 __gnu_pbds::tree_order_statistics_node_update>;
```

1.5. Custom Hash

While this slows down unordered_map a little, it helps prevent hacks.

```
1 struct custom_hash { // PERF: https://codeforces.com/blog/entry/62393
2     static uint64_t spltmix64(uint64_t x) {
3         x += 0x9e3779b97f4a7c15;
4         x = (x^(x>>30)) * 0xbf58476d1ce4e5b9;
5         x = (x^(x>>27)) * 0x94d049bb133111eb;
6         return x^(x>>31);
7     }
8     size_t operator()(uint64_t x) const {
9         static const uint64_t FIXED_RANDOM
10             = chrono::steady_clock::now().time_since_epoch().count();
11         return spltmix64(x + FIXED_RANDOM);
12     }
13 };
```

1.6. Random Number Generation

To generate a random integer, first set up gen.

```
mt19937_64 gen(chrono::steady_clock::now().time_since_epoch().count());
```

After which, you can use uniform_int_distribution to produce a random integer in a closed interval.

2. Disjoint Set Union

Nodes with negative values are roots. cmpCnt and cmpMax find the number of components and the size of the largest component respectively. Omit if not necessary.

```
1 struct DSU {
2     int cmpCnt, cmpMax;
3     V<int> nodes; // WARN: assumes `nodes` is zero-indexed
4     DSU(int n)
5         : cmpCnt(n)
6         , cmpMax(1)
7         , nodes(n, -1) {}
8     int find(int x) { return nodes[x] < 0 ? x : nodes[x] = find(nodes[x]); }
9     int size(int x) { return -nodes[find(x)]; }
10    bool unite(int x, int y) { // WARN: merge `y` into `x`
11        x = find(x), y = find(y);
12        if (x == y) return false;
13        if (nodes[x] > nodes[y]) swap(x, y);
14        nodes[x] += nodes[y], nodes[y] = x;
15        cmpCnt--;
16        ckmx(cmpMax, -nodes[x]);
17        return true;
18    }
19 };
```

2.1. Persistent Variant

Check if two nodes are connected at some point in time.

```
1 struct DSU {
2     V<int> size, parent, time_changed; // WARN: assumes `parent` is zero-indexed
3     DSU(int n)
4         : size(n, 1)
5         , parent(n, 0)
6         , time_changed(n, 0) {
7         iota(A(parent), 0);
8     }
9     int find(int x, int time) {
10        if (parent[x] == x || time_changed[x] > time) return x;
11        return find(parent[x], time);
12    }
13    bool unite(int x, int y, int time) { // WARN: merge `y` into `x`
14        x = find(x, time), y = find(y, time);
15        if (x == y) return false;
16        if (size[x] < size[y]) swap(x, y);
17        size[x] += size[y], parent[y] = x, time_changed[y] = time;
18        return true;
19    }
20 };
```

2.2. Kruskal Reconstruction Tree

This data structure stores the implicit tree formed across unite operations. kweight stores the weight of the edge that was used in unite. ops is the total number of operations. Even if x and y are connected, we would just add a single edge in the KRT. The eventual root will be kidx, which you may use to run the DFS to solve.

```
1 struct DSU {
2     int kidx;
3     vector<int> nodes, krep, kweight; // WARN: assumes `nodes` is zero-indexed
4     vector<vector<int>> krt;
5     DSU(int n, int ops)
6         : kidx(n - 1)
7         , nodes(n, -1)
8         , krep(n + ops, 0)
9         , kweight(n + ops, 0)
10        , krt(n + ops) {
11        iota(A(krep), 0);
12    }
13    int find(int x) { return nodes[x] < 0 ? x : nodes[x] = find(nodes[x]); }
14    int size(int x) { return -nodes[find(x)]; }
15    bool unite(int x, int y, int w) { // WARN: merge `y` into `x`
16        x = find(x), y = find(y);
17        if (x == y) {
18            kidx++;
19            krt[kidx].P(krep[x]);
20            kweight[kidx] = w;
21            krep[x] = kidx;
22            return false;
23        }
24        if (nodes[x] > nodes[y]) swap(x, y);
25        nodes[x] += nodes[y], nodes[y] = x;
26        kidx++;
27        krt[kidx].P(krep[x]);
28        krt[kidx].P(krep[y]);
29        kweight[kidx] = w;
30        krep[x] = kidx;
31        return true;
32    }
33    void solve(int u, int prev_w = oo) {
34        int min_w = min(prev_w, kweight[u]);
35        for (auto v : krt[u]) {
36            solve(v, min_w);
37        }
38    }
39 };
```

3. Sorting

3.1. Custom struct/class

Suppose we have some custom struct representing fractions, to create a set, we need to perform operator overloading of <.

```
1 struct Frac {
2   int x, y;
3   Frac(int x, int y)
4     : x(x)
5     , y(y) {}
6   bool operator<(Frac other) const { return x * other.y < y * other.x; }
7   };
```

3.2. Inversions Count

An array’s *inversions count* refers to the number of adjacent swaps required to sort an array. It can be computed via a modified merge sort, which has the side effect of sorting the array.

```
1 auto merge = [](auto self, V<int> &v, int lb, int rb) -> ll {
2   if (lb == rb) return 0;
3   int mb = (lb + rb) / 2, lp = lb, rp = mb + 1;
4   ll ans = self(self, v, lb, mb) + self(self, v, mb + 1, rb);
5   while (lp <= mb && rp <= rb) {
6     if (v[lp] <= v[rp]) lp++;
7     else ans += mb - lp + 1, rp++;
8   }
9   inplace_merge(B(v) + lb, B(v) + mb + 1, B(v) + rb + 1);
10  return ans;
11 };
```

4. Matching

4.1. Maximum Cardinality Bipartite Matching (MCMF)

A maximum matching contains as many edges of a graph G as possible such that each vertex is part of at most one edge of the subset. Assume that $0 \leq u < lsize$ and $0 \leq v < rsize$ in the following implementation.

```
1 int l_size, r_size, edge_cnt;
2 mt19937 gen(chrono::steady_clock::now().time_since_epoch().count());
3
4 int main() {
5   cin >> l_size >> r_size >> edge_cnt;
6   V<ii> edg, matching;
7
8   // WARN: assumes both `u` and `v` are zero-indexed
9   for (int i = 0, u, v; i < edge_cnt; i++) {
10    cin >> u >> v;
11    edg.push_back({u, v});
12  }
13
14  shuffle(A(edg), gen);
15
16  auto hopcroft_karp = [&]() -> void {
17    V<int> deg(l_size + 1);
18    for (auto &[u, v] : edg) deg[u]++;
19    partial_sum(begin(deg), end(deg), begin(deg));
20    V<int> g(edge_cnt), lmc(l_size, -1), rmc(r_size, -1), a, p, q(l_size);
21    for (auto &[u, v] : edg) g[--deg[u]] = v;
22    while (true) {
23      a.assign(l_size, -1), p.assign(l_size, -1);
24      int t = 0, match = false;
25      for (int i = 0; i < l_size; i++) {
26        if (lmc[i] == -1) q[t++] = a[i] = p[i] = i;
27      }
28      for (int i = 0; i < t; i++) {
29        int x = q[i];
30        if (!lmc[a[x]]) continue;
31        for (int j = deg[x]; j < deg[x + 1]; j++) {
32          int y = g[j];
33          if (rmc[y] == -1) {
34            while (--y) {
35              rmc[y] = x, swap(lmc[x], y), x = p[y];
36            }
37            match = true;
38            break;
39          }
40          if (p[rmc[y]] == -1) {
41            q[t++] = y = rmc[y], p[y] = x, a[y] = a[x];
```

```
42    }
43    }
44    }
45    if (!match) break;
46    }
47    for (int i = 0; i < l_size; i++) {
48      if (!lmc[i]) matching.push_back({i, lmc[i]});
49    }
50  };
51 }
```

4.1.1. Minimum Vertex Cover

Assuming you have obtained the matching above, you can retrieve the *minimum vertex cover* (the smallest set of vertices that includes at least one endpoint of every edge of the graph) by constructing a new *directed* graph. Edges that belong to the matching will go from right to left, whereas all other edges will go from left to right. Perform a depth first search starting at all left vertices that are not incident to any edges in the matching.

The minimum vertex cover includes all visited right vertices of the matching, as well as all unvisited left vertices of the matching.

Note that some problems also have situations where there are isolated nodes (degree zero) that have to be taken.

```
1 int l_size, r_size;
2 V<ii> matching, edg;
3 V<int> lhs, rhs;
4
5 int main() {
6   set<ii> seen;
7   set<ii> matching_set(A(matching));
8
9   map<ii, V<ii>> adj;
10  set<int> to_start;
11  V<int> min_cover;
12
13  F(i, 0, l_size) to_start.I(i);
14
15  for (auto [u, v] : edg) {
16    if (matching_set.C({u, v})) {
17      adj[{v, 1}].P({u, 0});
18      to_start.X(u);
19    } else {
20      adj[{u, 0}].P({v, 1});
21    }
22  }
23
24  auto dfs = [&](auto self, int u, int state) -> void {
25    seen.I({u, state});
26    for (auto [v, nstate] : adj[{u, state}]) {
27      if (seen.C({v, nstate})) continue;
28      self(self, v, nstate);
29    }
30  };
31
32  for (auto u : to_start) {
33    if (seen.C({u, 0})) continue;
34    dfs(dfs, u, 0);
35  }
36
37  for (auto [u, v] : matching) {
38    if (seen.C({v, 1})) {
39      min_cover.P(rhs[v]);
40    }
41    if (!seen.C({u, 0})) {
42      min_cover.P(lhs[u]);
43    }
44  }
45 }
```

4.2. Perfect Matching

A perfect matching is one where every vertex is included in exactly one edge of the matching. The following implementation finds the minimum weight perfect matching. To obtain the maximum weight variant, simply invert all the costs.

```
1 template <typename T>
2 class Hungarian {
3   const T INF = numeric_limits<T>::max();
4
5 public:
6   T tot = 0;
7   vector<int> p, way, ans;
8   vector<T> u, v;
9   Hungarian(const vector<vector<T>> &a) { // WARN: assumes `a` is one-indexed
10    int n = ssize(a) - 1, m = ssize(a[0]) - 1;
11    u.assign(n + 1, 0), v.assign(m + 1, 0);
12    p.assign(m + 1, 0), way.assign(m + 1, 0), ans.assign(n + 1, 0);
13    for (int i = 1; i <= n; i++) {
14      p[0] = i;
15      int j0 = 0;
16      vector<bool> usd(m + 1);
17      vector<T> mnv(m + 1, INF);
18      do {
19        usd[j0] = true;
20        int i0 = p[j0], j1 = 0;
21        T dlt = INF;
22        for (int j = 1; j <= m; j++) {
23          if (!usd[j]) {
24            T cur = a[i0][j] - u[i0] - v[j];
25            if (cur < mnv[j]) mnv[j] = cur, way[j] = j0;
26            if (mnv[j] < dlt) dlt = mnv[j], j1 = j;
27          }
28        }
29        for (int j = 0; j <= m; j++) {
30          if (usd[j]) u[p[j]] += dlt, v[j] -= dlt;
31          else mnv[j] -= dlt;
32        }
33        j0 = j1;
34      } while (p[j0]);
35      do {
36        int j1 = way[j0];
37        p[j0] = p[j1];
38        j0 = j1;
39      } while (j0);
40    }
41    for (int j = 1; j <= m; j++) {
42      ans[p[j]] = j;
43    }
44    tot = -v[0];
45  }
46  };
```

5. Square Root Decomposition

Assuming $block$ represents the size of a chunk, a represents the original, full-sized array, and b represents the values of each block, the example snippet demonstrates a sum query. A good value for $block$ is around 250.

```
1 // WARN: assumes zero-indexed queries
2 auto query = [](int l, int r) -> ll {
3   ll ans = 0;
4   int bl = l / block, br = r / block;
5   if (bl == br) return reduce(begin(a) + l, begin(a) + r + 1);
6   for (int i = l; i <= (bl + 1) * block - 1; i++) ans += a[i];
7   for (int i = bl + 1; i <= br - 1; i++) ans += b[i];
8   for (int i = br * block; i <= r; i++) ans += a[i];
9   return ans;
10 };
```

5.1. Mo’s Algorithm

```
1 // WARN: assumes zero-indexed queries
2 struct Query {
3   int i, l, r;
```

```

4   Query(int i, int l, int r)
5   : i(i)
6   , l(l)
7   , r(r) {}
8   friend bool operator<(const Query &q, const Query &qb) {
9       if (qa.l / block == qb.l / block) return qa.r < qb.r;
10      return qa.l < qb.l;
11  }
12 };
13 V<Query> queries;
14
15 void solve() {
16     auto ins = [&](int i) {};
17     auto del = [&](int i) {};
18
19     int cl = 0, cr = -1, cur = 0;
20
21     for (auto [i, l, r] : queries) {
22         while (cl > l) ins(--cl);
23         while (cr < r) ins(++cr);
24         while (cl < l) del(cl++);
25         while (cr > r) del(cr--);
26         ans[i] = cur;
27     }
28 }

```

6. Treap

6.1. Implicit Variant

Insertion and deletion are done based on the zero-indexed position of an element within the tree. Nodes in the snippet store string values, remember to modify depending on the problem.

```

1   mt19937_64 gen(chrono::steady_clock::now().time_since_epoch().count());
2   uniform_int_distribution dist(0, oo);
3
4   struct Node {
5       string val;
6       int pri, size;
7       Node *lp, *rp;
8       Node(string val)
9           : val(val)
10          , pri(dist(gen))
11          , size(1)
12          , lp(nullptr)
13          , rp(nullptr) {}
14 };
15
16 int size(Node *root) { return root ? root->size : 0; }
17
18 void split(Node *root, Node *&lt, Node *&rt, int idx) {
19     if (!root) {
20         lt = rt = nullptr;
21         return;
22     }
23     if (size(root->lp) + 1 <= idx) {
24         split(root->rp, root->rp, rt, idx - (size(root->lp) + 1));
25         lt = root;
26     } else {
27         split(root->lp, lt, root->lp, idx);
28         rt = root;
29     }
30     root->size = 1 + size(root->lp) + size(root->rp);
31 }
32
33 void merge(Node *&root, Node *lt, Node *rt) {
34     if (!lt || !rt) {
35         root = lt ? lt : rt;
36         return;
37     }
38     if (lt->pri > rt->pri) {
39         merge(lt->rp, lt->rp, rt);
40         root = lt;
41     } else {
42         merge(rt->lp, lt, rt->lp);
43         root = rt;

```

```

44 }
45 root->size = 1 + size(root->lp) + size(root->rp);
46 }
47
48 void concat(Node *root, vector<string> &ans) {
49     if (!root) return;
50     concat(root->lp, ans);
51     ans.push_back(root->val);
52     concat(root->rp, ans);
53 }
54
55 void clean(Node *root) {
56     if (!root) return;
57     clean(root->lp);
58     clean(root->rp);
59     delete root;
60 }
61
62 int n, q, find_idx = -1;
63
64 void find(Node *root, string &s, int offset) {
65     if (!root) return;
66
67     if (root->val == s) {
68         int cur = offset;
69
70         if (root->lp) cur += root->lp->size;
71         find_idx = cur;
72     }
73     return;
74 }
75
76 int inc = 1;
77 if (root->lp) inc += root->lp->size;
78
79 find(root->lp, s, offset);
80 find(root->rp, s, offset + inc);
81 }
82
83 void solve_t() {
84     cin >> n >> q;
85
86     vector<string> a(n);
87
88     F(i, 0, n) cin >> a[i];
89
90     Node *root = new Node(a[0]);
91
92     auto insert = [&](int i, string s) -> void {
93         Node *lp, *rp;
94         split(root, lp, rp, i);
95         Node *n_node = new Node(s);
96         merge(lp, lp, n_node);
97         merge(lp, lp, rp);
98         root = lp;
99     };
100
101     auto erase = [&](int i) -> void {
102         Node *lp, *rp, *n_lp, *n_rp;
103
104         split(root, lp, rp, i);
105         split(rp, n_lp, n_rp, 1);
106
107         V<string> ans;
108
109         merge(lp, lp, n_rp);
110         root = lp;
111     };
112
113     clean(root);
114 }

```

6.2. Cartesian Tree

We may also construct it from an array, assuming no further updates to the array are required. The smallest value belongs to the root, which stores the corresponding index. Assuming all values are distinct, the resulting tree is unique.

```

1   template <typename T>
2   class CartesianTree {
3   struct Node {
4       int p, l, r;
5       Node(int p = 0, int l = 0, int r = 0)
6           : p(p)
7           , l(l)
8           , r(r) {}
9   };
10
11 public:
12     int root;
13     deque<Node> tree;
14     CartesianTree(const vector<T> &a)
15         : tree(S(a)) {
16         for (int i = 1; i <= S(a) - 1; i++) { // WARN: 'a' is one-indexed
17             tree[i].p = i - 1;
18             while (a[i] < a[tree[i].p]) tree[i].p = tree[tree[i].p].p;
19             tree[i].l = tree[tree[i].p].r;
20             tree[tree[i].l].p = i;
21             tree[tree[i].p].r = i;
22         }
23         for (int i = 1; i <= S(a) - 1; i++)
24             if (tree[i].p == 0) root = i;
25     }
26 };

```

6.3. Lazy Propagation

The following snippet illustrates a non-implicit treap with range addition.

```

1   mt19937 gen(chrono::steady_clock::now().time_since_epoch().count());
2   uniform_int_distribution dist(0, INT_MAX);
3
4   struct Node {
5       int val, pri, add, q; // WARN: rmb to tweak (may not need `q`)
6       Node *lp, *rp;
7       Node(int val, int q)
8           : val(val)
9           , pri(dist(gen))
10          , add(0)
11          , q(q)
12          , lp(0)
13          , rp(0) {}
14 };
15
16 void push(Node *root) {
17     if (!root) return;
18     if (root->add) {
19         if (root->lp) root->lp->val += root->add, root->lp->add += root->add;
20         if (root->rp) root->rp->val += root->add, root->rp->add += root->add;
21         root->add = 0;
22     }
23 }
24
25 void split(Node *root, Node *&lt, Node *&rt, int x) {
26     if (!root) {
27         lt = rt = nullptr;
28         return;
29     }
30     push(root), push(lt), push(rt);
31     if (root->val <= x) {
32         split(root->rp, root->rp, rt, x);
33         lt = root;
34     } else {
35         split(root->lp, lt, root->lp, x);
36         rt = root;
37     }
38 }
39
40 void merge(Node *&root, Node *lt, Node *rt) {
41     push(root), push(lt), push(rt);
42     if (!lt || !rt) {
43         root = lt ? lt : rt;
44         return;
45     }
46     if (lt->pri >= rt->pri) {
47         merge(lt->rp, lt->rp, rt);
48         root = lt;

```

```
49 } else {
50     merge(rt->lp, lt, rt->lp);
51     root = rt;
52 }
53 }
54
55 void range_add(Node *root, int l, int r, int addend) {
56     Node *lt = nullptr, *lm = nullptr, *mt = nullptr, *rt = nullptr;
57     split(root, lm, rt, r);
58     split(lm, lt, mt, l - 1);
59     if (mt) mt->val += addend, mt->add += addend;
60     merge(lm, lt, mt);
61     merge(root, lm, rt);
62 }
63
64 void insert(Node *root, int x, int q) {
65     push(root);
66     Node *lt = nullptr, *rt = nullptr;
67     split(root, lt, rt, x);
68     merge(lt, lt, new Node(x, q));
69     merge(root, lt, rt);
70 }
71
72 void concat(Node *root, vector<ii> &a) {
73     if (!root) return;
74     push(root);
75     concat(root->lp, a);
76     a.push_back({ root->q, root->val });
77     concat(root->rp, a);
78 }
79
80 void clean(Node *root) {
81     if (!root) return;
82     clean(root->lp);
83     clean(root->rp);
84     delete root;
85 }
```

7. Graphs

7.1. Cycle Retrieval

In problems where $a[i]$ represents the next “jump” from position i , the following snippet finds all cycles, which may be used for further processing.

```
1 int main() {
2     V seen(n + 1), a(n + 1);
3
4     G(i, 1, n) cin >> a[i];
5
6     // WARN: assumes one-indexed `a`
7     for (int i = 1, j, cyces; i <= n; i++) {
8         if (seen[i]) continue;
9         j = i, cycs = 0;
10        set<int> curs;
11        deque<int> path;
12        do {
13            seen[j] = true, curs.insert(j), path.push_back(j);
14            j = a[j];
15            if (curs.contains(j)) {
16                cycs = j;
17                break;
18            }
19        } while (!seen[j]);
20        if (cyces) {
21            while (path.front() != cycs) path.pop_front();
22        }
23    }
24 }
```

7.2. Diameter

The furthest distance a node u is from some other node in a tree is equivalent to $\max(\text{dist}(\text{dia}_u, u), \text{dist}(\text{dia}_v, v))$, where dia_u and dia_v represent the endpoints of some diameter.

```
1 // WARN: assumes zero-indexed `adj`
2 auto build = [](vector<vector<int>> &adj) -> pair<int, vector<int>> {
3     int init = 0, n = S(adj);
4     V<int> df(n, 0), du(n, 0), dv(n, 0);
5
6     auto dfs
7     = [&](auto &&self, int u, int pu, vector<int> &d, int cur = 0) -> void {
8         d[u] = cur;
9         for (auto v : adj[u]) {
10             if (v == pu) continue;
11             self(self, v, u, d, cur + 1);
12         }
13     };
14
15     dfs(dfs, init, -1, df);
16     int dia_u = max_element(A(df)) - begin(df);
17
18     dfs(dfs, dia_u, -1, du);
19     int dia_v = max_element(A(du)) - begin(du);
20
21     dfs(dfs, dia_v, -1, dv);
22     vector<int> ans(n, 0);
23
24     F(i, 0, n) { ckmx(ans[i], max(du[i], dv[i])); }
25
26     return { *max_element(A(du)), ans };
27 };
```

7.3. Bridges

A *bridge* is an edge of a graph whose deletion increases the graph’s number of connected components. $\text{memo}[u]$ represents the number of *back-edges* passing over the edge between u and its parent.

```
1 int n;
2
3 int main() {
4     cin >> n;
5
6     V adj(n, V());
7     vector<ii> bridges;
8     vector<int> depth(n), memo(n);
9
10    // NOTE: assumes zero-indexed nodes
11    auto solve = [&](auto self, int u, int pu) -> void {
12        for (auto &v : adj[u])
13            if (v != pu) {
14                if (depth[v]) {
15                    depth[v] = depth[u] + 1;
16                    self(self, v, u);
17                    memo[u] += memo[v];
18                } else if (depth[v] < depth[u]) {
19                    memo[u]++;
20                } else {
21                    memo[u]--;
22                }
23            }
24            if (depth[u] != 1 && !memo[u]) bridges.push_back({ pu, u });
25        };
26
27        depth[0] = 1;
28        solve(solve, 0, -1);
29    }
```

7.4. Cut Vertices

When a *cut vertex* is removed, the graph is disconnected.

```
1 int n;
2
3 int main() {
4     cin >> n;
5
6     V adj(n, V());
7     vector<int> depth(n), memo(n, INT_MAX), cuts(n, false);
8
9     // NOTE: assumes zero-indexed nodes
```

```
10 auto solve = [&](auto self, int u, int pu) -> void {
11     int children = 0;
12     for (auto &v : adj[u])
13         if (v != pu) {
14             if (depth[v]) {
15                 depth[v] = depth[u] + 1;
16                 self(self, v, u);
17                 memo[u] = min(memo[u], memo[v]);
18                 children++;
19             } else if (depth[v] < depth[u]) {
20                 memo[u] = min(memo[u], depth[v]);
21             } else {
22                 memo[v] = min(memo[v], depth[u]);
23             }
24         }
25     if (pu == -1) {
26         cuts[u] = (children >= 2);
27     } else {
28         if (memo[u] >= depth[pu]) cuts[pu] = true;
29     }
30 };
31
32 depth[0] = 1;
33 solve(solve, 0, -1);
34 }
```

7.5. Edge Cactus

Every edge belongs to at most one cycle. You may find all cycles by merging the edges with DSU.

```
1 int n, m;
2
3 int main() {
4     cin >> n >> m;
5
6     V adj(n + 1, V<ii>());
7     V<ii> edg;
8
9     F(i, 0, m) {
10        int u, v;
11        cin >> u >> v;
12        adj[u].P({ v, i });
13        adj[v].P({ u, i });
14        edg.P({ u, v });
15    };
16
17    V mnb(n + 1, n + 1), depth(n + 1, 0);
18    mnb[n] = n;
19    DSU dsu(m);
20
21    auto solve = [&](auto &&self, int u, int pu, int ei) -> ii {
22        depth[u] = depth[pu] + 1;
23        ii dmn = { depth[u], ei };
24        for (auto [v, en] : adj[u]) {
25            if (v == pu) continue;
26            if (depth[v]) ckmn(dmn, ii{ depth[v], en });
27            else ckmn(dmn, (ii)self(self, v, u, en));
28        }
29        if (dmn[0] != depth[u]) {
30            int en = dmn[1];
31            dsu.unite(ei, en);
32        }
33        return dmn;
34    };
35
36    G(i, 1, n) {
37        if (depth[i]) continue;
38        solve(solve, i, 0, -1);
39    }
40 }
```

7.6. Eulerian Paths/Circuits

7.6.1. Undirected

Assuming we have an undirected *connected* graph, if all vertices have an *even* degree, an Eulerian circuit exists; otherwise, if we have two vertices of odd degree, an Eulerian path exists (o1 and o2 represent the two odd vertices).

Be careful about self-loops, they don't count towards the degree! In general this algorithm is pretty expensive, so perform it *just once* if possible.

```
1 int odds = 0, n, m, o1, o2;
2 V adj(n + 1, set<int>());
3 V p;
4
5 int main() {
6     // NOTE: do some pre-processing
7     // ...
8
9     if (odds == 1 || odds >= 3) {
10         cout << "IMPOSSIBLE" << "\n";
11         return 0;
12     }
13
14     auto solve = [&](auto self, int u) -> void {
15         auto &vs = adj[u];
16         while (!empty(vs)) {
17             int v = *begin(vs);
18             vs.erase(v);
19             adj[v].erase(u);
20             self(self, v);
21         }
22         p.push_back(u);
23     };
24     // WARN: assume one-indexed nodes
25     solve(solve, odds == 2 ? o1 : 1);
26
27     if (S(p) != m + 1) {
28         cout << "IMPOSSIBLE" << "\n";
29         return 0;
30     }
31     for (int i = 0; i < m + 1; i++) cout << p[i] << " \n"[i == m];
32 }
```

7.6.2. Directed

A directed graph has an Eulerian circuit if and only if it is *connected* and each vertex has the same in-degree as out-degree. A directed graph has an Eulerian path if and only if it is connected and each vertex except two has an in-degree as out-degree.

One of the two should have an out-degree one greater than in-degree (*start vertex*), while the other has an in-degree one greater than out-degree (*end vertex*). The core algorithm is the same as that for the undirected graph.

7.7. Binary Lifting

Before we can perform binary lifting, we have to first construct a table `tab`, where `tab[i][h]` represents the 2^h -th ancestor of `i`.

```
1 int n;
2
3 int main() {
4     // WARN: assumes zero-indexed nodes
5     V adj(n, V());
6
7     auto ceil_log2 = [](int n) -> int {
8         int ans = 0, cur = 1;
9         while (cur < n) cur *= 2, ans++;
10        return ans;
11    };
12
13    int tab_height = ceil_log2(n);
14    vector depth(n, 0);
```

```
15    vector tab(n, vector(tab_height + 1, -1));
16
17    auto build_tab = [&](auto self, int u, int pu) -> void {
18        tab[u][0] = pu;
19        for (auto &v : adj[u])
20            if (v != pu) {
21                depth[v] = depth[u] + 1;
22                self(self, v, u);
23            }
24    };
25    depth[0] = 1;
26    build_tab(build_tab, 0, -1);
27
28    for (int k = 1; k <= tab_height; k++) {
29        for (int u = 0; u < n; u++) {
30            if (tab[u][k - 1] != -1) {
31                tab[u][k] = tab[tab[u][k - 1]][k - 1];
32            }
33        }
34    }
35
36    auto get_parent = [&](int u, int k) -> int {
37        for (int h = 0; h <= tab_height; h++)
38            if ((k >> h) & 1) {
39                u = tab[u][h];
40                if (u == -1) break;
41            }
42        return u;
43    };
44
45    auto get_lca = [&](auto self, int u, int v) -> int {
46        if (depth[u] < depth[v]) return self(v, u);
47        u = get_parent(u, depth[u] - depth[v]);
48        if (u == v) return u;
49        for (int h = tab_height; h >= 0; h--) {
50            if (tab[u][h] != tab[v][h]) u = tab[u][h], v = tab[v][h];
51        }
52        return tab[u][0];
53    };
54
55    auto get_dist = [&](auto self, int u, int v) -> int {
56        if (depth[u] < depth[v]) return self(v, u);
57        int delta = depth[u] - depth[v], ans = delta;
58        u = get_parent(u, delta);
59        if (u == v) return delta;
60        for (int h = tab_height; h >= 0; h--) {
61            if (tab[u][h] != tab[v][h]) {
62                u = tab[u][h], v = tab[v][h];
63                ans += 2 * (1 << h);
64            }
65        }
66        return ans + 2;
67    };
68 }
```

7.8. Heavy Light Decomposition

If you want to update the value of node `u`, you need to update `pos[u]`. query needs to be tweaked depending on the problem.

```
1 int n;
2 constexpr int NMAX = 100'000;
3
4 int main() {
5     // WARN: assumes zero-indexed nodes
6     V adj(n, V());
7
8     V raw(n), s(n); // NOTE: 's' contains the value in each node
9     array<int, NMAX> par{}, depth{}, heavy{}, head{}, pos{};
10    fill(A(heavy), -1);
11    int cur_pos = 0;
12
13    auto dfs = [&](auto self, int u) -> int {
14        int subtree = 1, max_child = 0;
15        for (auto &v : adj[u])
16            if (par[u] != v) {
17                par[v] = u, depth[v] = depth[u] + 1;
18                int child = self(self, v);
19                if (child > max_child) max_child = child, heavy[u] = v;
```

```
20    }
21    return subtree;
22 };
23 dfs(dfs, 0);
24
25 auto dc = [&](auto self, int u, int h) -> void {
26     head[u] = h, pos[u] = cur_pos, raw[cur_pos] = s[u], cur_pos++;
27     if (heavy[u] != -1) self(self, heavy[u], h);
28     for (auto &v : adj[u])
29         if (par[u] != v && v != heavy[u]) {
30             self(self, v, v);
31         }
32 };
33 dc(dc, 0, 0);
34
35 SegTree tree(raw);
36 auto query = [&](int u, int v) -> int {
37     int ans = numeric_limits<int>::min();
38     for (; head[u] != head[v]; v = par[head[v]]) {
39         if (depth[head[u]] > depth[head[v]]) swap(u, v);
40         ans = tree.conquer(ans, tree.query(pos[head[v]], pos[v]));
41     }
42     if (depth[u] > depth[v]) swap(u, v);
43     return tree.conquer(ans, tree.query(pos[u], pos[v]));
44 };
45 }
```

7.9. Tree Hashing

You can hash a *rooted* tree like so. If it isn't rooted, you may find a centroid first before using it as a root to perform the hash. Some trees may have *two* centroids, in which case you have to retrieve two hashes. Among the two, put the smaller one before the larger one in a pair of integers (e.g., in `V<int>`).

```
1 auto hash = [](vector<vector<int>>& adj) -> array<Mint, 2> {
2     vector subs(n + 1, Mint());
3
4     // WARN: assumes one-indexed nodes
5     auto solve
6         = [&](auto self, int b = 31, int u = 1, int pu = 0, int d = 0) -> Mint {
7         deque<Mint> vs;
8         subs[u]++;
9
10        for (auto v : adj[u]) {
11            if (v == pu) continue;
12            vs.P(subs[v] * self(self, b, v, u, d + 1));
13            subs[u] += subs[v];
14        }
15        sort(A(vs), [&](Mint &x, Mint &y) { return x.v < y.v; });
16
17        vs.push_front(d), vs.push_back(d);
18
19        Mint ans = 0;
20
21        F(i, 0, S(vs)) { ans += vs[i] * Mint(b).pow(i); }
22
23        return ans;
24    };
25
26    return { solve(solve, 31), solve(solve, 39) };
27 };
```

7.10. Centroid Decomposition

`ancestors[u]` returns the list of ancestors of `u` (along with the distance) as we go up the centroid decomposition tree.

```
1 int main() {
2     // WARN: assumes one-indexed nodes
3     cin >> n;
4     V adj(n + 1, V());
5
6     F(_, 0, n - 1) {
7         int u, v;
8         cin >> u >> v;
9         adj[u].P(v);
```



```
10 adj[v].P(u);
11 }
12
13 V seen(n + 1), subt(n + 1, 0);
14 V ancestors(n + 1, V<ii>());
15
16 auto build_subt = [&](auto self, int u, int pu) -> int {
17     subt[u] = 1;
18     for (auto v : adj[u]) {
19         if (v == pu || seen[v]) continue;
20         subt[u] += self(self, v, u);
21     }
22     return subt[u];
23 };
24
25 // WARN: returns one centroid only (there might be two)
26 auto get_centroid = [&](auto self, int u, int pu, int tree_size) -> int {
27     for (auto v : adj[u]) {
28         if (v == pu || seen[v]) continue;
29         if (2 * subt[v] > tree_size) return self(self, v, u, tree_size);
30     }
31     return u;
32 };
33
34 auto build_ancestor
35     = [&](auto self, int u, int pu, int centroid, int cur_d = 1) -> void {
36     for (auto v : adj[u]) {
37         if (v == pu || seen[v]) continue;
38         self(self, v, u, centroid, cur_d + 1);
39     }
40     ancestors[u].P({ centroid, cur_d });
41 };
42
43 auto centroid_decomp = [&](auto self, int init = 1) -> void {
44     int centroid
45     = get_centroid(get_centroid, init, 0, build_subt(build_subt, init, 0));
46     for (auto v : adj[centroid]) {
47         if (seen[v]) continue;
48         build_ancestor(build_ancestor, v, centroid, centroid);
49     }
50     seen[centroid] = true;
51     for (auto v : adj[centroid]) {
52         if (seen[v]) continue;
53         self(self, v);
54     }
55 };
56 }
```

7.11. 2-SAT

```
1 // WARN: assumes zero-indexed nodes
2 class two_sat {
3     int n;
4     vector<vector<int>>> ori, rev;
5
6 public:
7     two_sat(int n)
8         : n(n) {
9         ori.assign(2 * n, {});
10        rev.assign(2 * n, {});
11    }
12    void add_clause(int u, bool u_pos, int v, bool v_pos) {
13        ori[2 * u + u_pos].push_back(2 * v + (v_pos));
14        ori[2 * v + v_pos].push_back(2 * u + (u_pos));
15        rev[2 * v + (v_pos)].push_back(2 * u + u_pos);
16        rev[2 * u + (u_pos)].push_back(2 * v + v_pos);
17    }
18    bool is_satisfiable() {
19        int color = 1;
20        stack<int> stk;
21        vector<int> seen(2 * n), colors(2 * n);
22        auto build_stk = [&](auto build_stk, int u) -> void {
23            seen[u] = true;
24            for (auto &v : ori[u])
25                if (!seen[v]) {
26                    build_stk(build_stk, v);
27                }
28            stk.push(u);
29        };
30    }
```

```
30     for (int u = 0; u < 2 * n; u++)
31         if (!seen[u]) build_stk(build_stk, u);
32     auto build_scc = [&](auto build_scc, int u) -> void {
33         colors[u] = color, seen[u] = true;
34         for (auto &v : rev[u])
35             if (!colors[v]) {
36                 build_scc(build_scc, v);
37             }
38     };
39     while (!stk.empty()) {
40         int u = stk.top();
41         stk.pop();
42         if (!colors[u]) {
43             build_scc(build_scc, u);
44             color++;
45         }
46     }
47     for (int i = 0; i < n; i++) {
48         if (colors[2 * i] == colors[2 * i + 1]) return false;
49     }
50     return true;
51 }
52 }
```

7.12. Small-To-Large Merging

Instead, of each node having its own set of objects, try to reuse the children sets as much as possible. This reduces the number of swaps/inserts and improves the constant factor. The following snippet solves CSES’s Distinct Colors, which asks you to find the number of distinct colors in each subtree.

```
1 void solve_t() {
2     cin >> n;
3
4     vector<set<int>>> col(n + 1);
5     vector adj(n + 1, vector<int>());
6     vector ans(n + 1, 0);
7     vector c(n + 1, 0);
8
9     G(u, 1, n) { cin >> c[u]; }
10
11     F(_, 0, n - 1) {
12         int u, v;
13         cin >> u >> v;
14         adj[u].P(v);
15         adj[v].P(u);
16     }
17
18     // auto merge = [&](set<int> &s, set<int> &t) -> void {
19     //     if (S(s) < S(t)) swap(s, t);
20     //     for (auto &c : t) s.l(c);
21     //     t.clear();
22     // };
23
24     auto solve = [&](auto self, int u = 1, int pu = 0) -> set<int> * {
25         vector<set<int>> * ptrs;
26
27         for (auto v : adj[u]) {
28             if (v == pu) continue;
29             ptrs.P(self(self, v, u));
30         }
31
32         if (empty(ptrs)) {
33             col[u].l(c[u]); ans[u] = 1;
34             return &col[u];
35         }
36
37         int mxi = 0, mxv = S(*ptrs[0]);
38
39         F(i, 1, S(ptrs)) {
40             if (S(*ptrs[i]) > mxv) {
41                 mxv = S(*ptrs[i]);
42                 mxi = i;
43             }
44         }
45
46         F(i, 0, S(ptrs)) {
47             if (i == mxi) continue;
```

```
48         for (auto x : *ptrs[i]) {
49             ptrs[mxi]->insert(x);
50         }
51         ptrs[i]->clear();
52     }
53
54     ptrs[mxi]->insert(c[u]);
55     ans[u] = S(*ptrs[mxi]);
56
57     return ptrs[mxi];
58 };
59 solve(solve);
60
61 G(u, 1, n) { cout << ans[u] << N(u, n); }
62 }
```

8. Flows

8.1. Edmonds-Karp

Suppose we have already set up `adj` and `cap`, where `cap` represents the capacity of each edge; we may use this algorithm to determine the maximum flow in a flow network. When adding an edge, we need to include both (u, v) and (v, u) in `adj`, however, we’ll only set the capacity of (u, v) to whatever it is.

To determine the **minimum cut**, find all vertices that can be reached from the source, using only edges with positive residual capacity.

```
1 int main() {
2     V adj(n, V()), cap(n, V(n, 0));
3
4     auto bfs = [&](int s, int t, vector<int> &par) -> int {
5         par.assign(ssize(adj), -1), par[s] = -2;
6         queue<i> q;
7         q.push({ s, oo });
8         while (!empty(q)) {
9             auto [u, flow] = q.front();
10            q.pop();
11            for (auto &v : adj[u]) {
12                if (par[v] != -1 || cap[u][v] == 0) continue;
13                par[v] = u;
14                int new_flow = min(flow, cap[u][v]);
15                if (v == t) return new_flow;
16                q.push({ v, new_flow });
17            }
18        }
19        return 0;
20    };
21
22    // WARN: you may need to alter `int` to `ll` depending on the max flow
23    auto max_flow = [&](int s, int t) -> int {
24        int flow = 0, new_flow = 0;
25        vector par(ssize(adj), 0);
26        while ((new_flow = bfs(s, t, par))) {
27            flow += new_flow;
28            int u = t;
29            while (u != s) {
30                int pu = par[u];
31                cap[pu][u] -= new_flow, cap[u][pu] += new_flow;
32                u = pu;
33            }
34        }
35        return flow;
36    };
37 }
```

8.2. Dinic’s Algorithm

Remember to reset all the edges if you need to compute the flow to some new destination. It’s also possible to get the incremental flow by adding edges and calling `flow` repeatedly.

```
1 struct FlowEdge {
```

```
2  int v, u;
3  ll cap, flow = 0;
4  FlowEdge(int v, int u, ll cap)
5      : v(v)
6      , u(u)
7      , cap(cap) {}
8  };
9
10 struct Dinic {
11     const ll flow_inf = OO;
12     vector<FlowEdge> edges;
13     vector<vector<int>>> adj;
14
15     int n, m = 0;
16     int s, t;
17
18     vector<int> level, ptr;
19     queue<int> q;
20
21     Dinic(int n, int s)
22         : n(n)
23         , s(s) {
24         adj.resize(n);
25         level.resize(n);
26         ptr.resize(n);
27     }
28
29     void add_edge(int v, int u, ll cap) { // NOTE: add edge (v -> u)
30         edges.emplace_back(v, u, cap);
31         edges.emplace_back(u, v, 0);
32         adj[v].push_back(m);
33         adj[u].push_back(m + 1);
34         m += 2;
35     }
36
37     bool bfs() {
38         while (!q.empty()) {
39             int v = q.front();
40             q.pop();
41             for (int id : adj[v]) {
42                 if (edges[id].cap == edges[id].flow) continue;
43                 if (level[edges[id].u] != -1) continue;
44                 level[edges[id].u] = level[v] + 1;
45                 q.push(edges[id].u);
46             }
47         }
48         return level[t] != -1;
49     }
50
51     ll dfs(int v, ll pushed) {
52         if (pushed == 0) return 0;
53         if (v == t) return pushed;
54         for (int &cid = ptr[v]; cid < (int)adj[v].size(); cid++) {
55             int id = adj[v][cid];
56             int u = edges[id].u;
57             if (level[v] + 1 != level[u]) continue;
58             ll tr = dfs(u, min(pushed, edges[id].cap - edges[id].flow));
59             if (tr == 0) continue;
60             edges[id].flow += tr;
61             edges[id ^ 1].flow -= tr;
62             return tr;
63         }
64         return 0;
65     }
66
67     ll flow(int t_) {
68         t = t_;
69         ll f = 0;
70         while (true) {
71             fill(A(level), -1);
72             level[s] = 0;
73             q.push(s);
74             if (!bfs()) break;
75             fill(A(ptr), 0);
76             while (ll pushed = dfs(s, flow_inf)) {
77                 f += pushed;
78             }
79         }
80         return f;
81     }
```

```
82  };
```

9. Sparse Table

```
1  template <typename T>
2  class SparseTable {
3      const int K = 25;
4      int n;
5      vector<vector<T>>> st;
6      T conquer(T x, T y) { return min(x, y); } // WARN: rmb to tweak
7  public:
8      SparseTable(vector<T> &a)
9          : n(S(a))
10            , st(K + 1, vector<T>(n)) {
11          copy(all(a), begin(st[0]));
12          for (int i = 1; i <= K; i++) {
13              for (int j = 0; j + (1 << i) <= n; j++) {
14                  st[i][j] = conquer(st[i - 1][j], st[i - 1][j + (1 << (i - 1))]);
15              }
16          }
17      }
18      T query(int l, int r) {
19          int i = __lg(r - l + 1);
20          return conquer(st[i][l], st[i][r - (1 << i) + 1]);
21      }
22  };
```

10. Fenwick Tree

10.1. One-Dimensional Queries

```
1  // NOTE: assumes one-indexed queries
2  template <typename T = int>
3  class FenwickTree {
4      int n;
5      vector<T> tree;
6
7  public:
8      FenwickTree(int tree_size)
9          : n(tree_size) {
10         tree.assign(n + 1, {});
11     }
12     void point_add(int i, T addend) {
13         for (; i <= n; i += Iso(i)) tree[i] += addend;
14     }
15     void range_add(int l, int r, T addend) {
16         point_add(l, addend);
17         point_add(r + 1, -addend);
18     }
19     T point_ask(int i) {
20         T sum = {};
21         for (; i := Iso(i)) sum += tree[i];
22         return sum;
23     }
24     T range_ask(int x, int y) { return point_ask(y) - point_ask(x - 1); }
25 };
```

10.2. Two-Dimensional Queries

```
1  // NOTE: assumes one-indexed queries
2  template <typename T = int>
3  class FenwickTree {
4      int height;
5      int width;
6      vector<vector<T>>> tree;
7
8  public:
9      FenwickTree(int h, int w)
10          : height(h)
11            , width(w) {
12         tree.assign(height + 1, vector<T>(width + 1));
13     }
14     T point_ask(int x, int y) {
15         T ans = {};
16         for (; x := Iso(x)) {
```

```
17         for (int i = y; i := Iso(i)) {
18             ans += tree[x][i];
19         }
20     }
21     return ans;
22 }
23 T area_ask(int x1, int y1, int x2, int y2) {
24     T ans = point_ask(x2, y2) + point_ask(x1 - 1, y1 - 1);
25     ans -= point_ask(x2, y1 - 1) + point_ask(x1 - 1, y2);
26     return ans;
27 }
28 void point_add(int x, int y, T addend) {
29     for (; x <= height; x += Iso(x)) {
30         for (int i = y; i <= width; i += Iso(i)) {
31             tree[x][i] += addend;
32         }
33     }
34 }
35 void area_add(int x1, int y1, int x2, int y2, T addend) {
36     point_add(x1, y1, addend);
37     point_add(x2 + 1, y2 + 1, addend);
38     point_add(x2 + 1, y1, -addend);
39     point_add(x1, y2 + 1, -addend);
40 }
41 };
```

10.3. Range Addition/Queries

This is a tiny bit faster compared to using segment trees.

```
1  // NOTE: assumes one-indexed queries
2  template <typename T = ll>
3  class FenwickTree {
4      int n;
5      vector<T> tree;
6
7  public:
8      FenwickTree(int tree_size)
9          : n(tree_size) {
10         tree.assign(n + 1, {});
11     }
12     void point_add(int i, T addend) {
13         for (; i <= n; i += Iso(i)) tree[i] += addend;
14     }
15     T point_ask(int i) {
16         T sum = {};
17         for (; i := Iso(i)) sum += tree[i];
18         return sum;
19     }
20 };
21 class DoubleFenwick {
22     int n;
23     FenwickTree<ll> b1, b2;
24     ll prefix_sum(int idx) { return b1.point_ask(idx) * idx - b2.point_ask(idx); }
25
26 public:
27     DoubleFenwick(int n)
28         : n(n)
29         , b1(n)
30         , b2(n) {}
31     void range_add(int l, int r, ll x) {
32         b1.point_add(l, x);
33         b1.point_add(r + 1, -x);
34         b2.point_add(l, x * (l - 1));
35         b2.point_add(r + 1, -x * r);
36     }
37     ll range_query(int l, int r) { return prefix_sum(r) - prefix_sum(l - 1); }
38 };
```

11. Segment Tree

11.1. RMQ (Iterative)

It can help to avoid TLE in certain situations.

```
1  template <typename T>
2  class SegTree {
```

```
3 int n;
4 vector<T> a, tree;
5 void build() {
6     for (int i = n - 1; i >= 1; --i) {
7         tree[i] = conquer(tree[i << 1], tree[(i << 1) ^ 1]);
8     }
9 }
10
11 public:
12 SegTree(vector<T> &a)
13     : a(a) {
14         n = ssize(a);
15         tree.assign(2 * n, {});
16         for (int i = 0; i < n; i++) tree[i + n] = a[i];
17         build();
18     }
19 T conquer(T x, T y) { // WARN: rmb to tweak
20     return min(x, y);
21 }
22 void update(int i, T val) {
23     for (tree[i += n] = val; i > 1; i >= 1) {
24         tree[i >> 1] = conquer(tree[i], tree[i ^ 1]);
25     }
26 }
27 T query(int l, int r) {
28     T ans = numeric_limits<T>::max(); // WARN: depends on `conquer`
29     for (l += n, r += n + 1; l < r; l >= 1, r >= 1) {
30         if (l & 1) ans = conquer(ans, tree[l++]);
31         if (r & 1) ans = conquer(ans, tree[--r]);
32     }
33     return ans;
34 }
35 };
```

11.2. RMQ (Recursive)

```
1 template <typename T>
2 class SegTree {
3     int n;
4     vector<T> a, tree;
5     void build(int i, int l, int r) {
6         if (l == r) tree[i] = a[l];
7         else {
8             int m = (l + r) / 2;
9             build(2 * i, l, m);
10            build(2 * i + 1, m + 1, r);
11            tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
12        }
13    }
14    void update(int i, int tl, int tr, int pos, T val) {
15        if (tl == tr) tree[i] = val;
16        else {
17            int tm = (tl + tr) / 2;
18            if (pos <= tm) update(2 * i, tl, tm, pos, val);
19            else update(2 * i + 1, tm + 1, tr, pos, val);
20            tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
21        }
22    }
23    T query(int i, int tl, int tr, int l, int r) {
24        if (l > r) return numeric_limits<T>::max(); // WARN: depends on `conquer`
25        if (l == tl && r == tr) return tree[i];
26        int tm = (tl + tr) / 2;
27        return conquer(
28            query(2 * i, tl, tm, l, min(r, tm)),
29            query(2 * i + 1, tm + 1, tr, max(l, tm + 1), r));
30    }
31 }
32 public:
33 SegTree(vector<T> &a)
34     : a(a) {
35         n = ssize(a);
36         tree.assign(4 * n, {});
37         build(1, 0, n - 1);
38     }
39 T conquer(T x, T y) { // WARN: rmb to tweak
40     return min(x, y);
41 }
42 void update(int pos, T val) { update(1, 0, n - 1, pos, val); }
43 T query(int l, int r) { return query(1, 0, n - 1, l, r); }
```

```
44 };
```

11.3. Lazy Propagation

11.3.1. Range Addition/Assignment (Min Query)

```
1 template <typename T>
2 class LazyTree {
3     struct LazyNode {
4         T assign_val;
5         T add_val;
6         bool has_assign;
7         bool has_add;
8         bool has_update() { return has_assign || has_add; }
9     };
10    int n;
11    vector<LazyNode> lazy;
12    vector<T> a, tree;
13    void build(int i, int l, int r) {
14        if (l == r) tree[i] = a[l];
15        else {
16            int m = (l + r) / 2;
17            build(2 * i, l, m);
18            build(2 * i + 1, m + 1, r);
19            tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
20        }
21    }
22    void push(int i) {
23        if (lazy[i].has_assign) {
24            tree[2 * i] = lazy[i].assign_val;
25            tree[2 * i + 1] = lazy[i].assign_val;
26            lazy[2 * i].add_val = lazy[2 * i + 1].add_val = {};
27            lazy[2 * i].has_add = lazy[2 * i + 1].has_add = false;
28            lazy[2 * i].assign_val = lazy[2 * i + 1].assign_val = lazy[i].assign_val;
29            lazy[2 * i].has_assign = lazy[2 * i + 1].has_assign = true;
30        } else {
31            if (lazy[2 * i].has_assign) {
32                tree[2 * i] += lazy[i].add_val;
33                lazy[2 * i].assign_val += lazy[i].add_val;
34            } else {
35                tree[2 * i] += lazy[i].add_val;
36                lazy[2 * i].add_val += lazy[i].add_val;
37                lazy[2 * i].has_add = true;
38            }
39            if (lazy[2 * i + 1].has_assign) {
40                tree[2 * i + 1] += lazy[i].add_val;
41                lazy[2 * i + 1].assign_val += lazy[i].add_val;
42            } else {
43                tree[2 * i + 1] += lazy[i].add_val;
44                lazy[2 * i + 1].add_val += lazy[i].add_val;
45                lazy[2 * i + 1].has_add = true;
46            }
47        }
48        lazy[i].has_assign = lazy[i].has_add = false;
49        lazy[i].assign_val = lazy[i].add_val = {};
50    }
51    void add(int i, int tl, int tr, int l, int r, T addend) {
52        if (l > r) return;
53        if (l == tl && r == tr) {
54            if (lazy[i].has_assign) {
55                tree[i] += addend;
56                lazy[i].assign_val += addend;
57            } else {
58                tree[i] += addend;
59                lazy[i].add_val += addend;
60                lazy[i].has_add = true;
61            }
62        } else {
63            if (lazy[i].has_update()) push(i);
64            int tm = (tl + tr) / 2;
65            add(2 * i, tl, tm, l, min(r, tm), addend);
66            add(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, addend);
67            tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
68        }
69    }
70    void assign(int i, int tl, int tr, int l, int r, T val) {
71        if (l > r) return;
72        if (l == tl && r == tr) {
73            if (lazy[i].has_add) {
```

```
74            lazy[i].has_add = false;
75            lazy[i].add_val = {};
76        }
77        tree[i] = val;
78        lazy[i].assign_val = val;
79        lazy[i].has_assign = true;
80    } else {
81        if (lazy[i].has_update()) push(i);
82        int tm = (tl + tr) / 2;
83        assign(2 * i, tl, tm, l, min(r, tm), val);
84        assign(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, val);
85        tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
86    }
87 }
88 T query(int i, int tl, int tr, int l, int r) {
89     if (l > r) return numeric_limits<T>::max(); // WARN: depends on `conquer`
90     if (l == tl && r == tr) return tree[i];
91     if (lazy[i].has_update()) push(i);
92     int tm = (tl + tr) / 2;
93     return conquer(
94         query(2 * i, tl, tm, l, min(r, tm)),
95         query(2 * i + 1, tm + 1, tr, max(l, tm + 1), r));
96 }
97
98 public:
99 LazyTree(vector<T> &a)
100     : a(a) {
101         n = ssize(a);
102         lazy.assign(4 * n, {});
103         tree.assign(4 * n, {});
104         build(1, 0, n - 1);
105     }
106 T conquer(T x, T y) { // WARN: rmb to tweak
107     return min(x, y);
108 }
109 void add(int l, int r, T addend) { add(1, 0, n - 1, l, r, addend); }
110 void assign(int l, int r, T val) { assign(1, 0, n - 1, l, r, val); }
111 T query(int l, int r) { return query(1, 0, n - 1, l, r); }
112 };
```

11.3.2. Range Addition/Assignment (Sum Query)

```
1 template <typename T>
2 class LazyTree {
3     struct LazyNode {
4         T assign_val;
5         T add_val;
6         bool has_assign;
7         bool has_add;
8         bool has_update() { return has_assign || has_add; }
9     };
10    int n;
11    vector<LazyNode> lazy;
12    vector<ll> s;
13    vector<T> a, tree;
14    void build(int i, int l, int r) {
15        if (l == r) tree[i] = a[l], s[i] = 1;
16        else {
17            int m = (l + r) / 2;
18            build(2 * i, l, m);
19            build(2 * i + 1, m + 1, r);
20            s[i] = (r - l + 1);
21            tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
22        }
23    }
24    void push(int i) {
25        if (lazy[i].has_assign) {
26            tree[2 * i] = s[2 * i] * lazy[i].assign_val;
27            tree[2 * i + 1] = s[2 * i + 1] * lazy[i].assign_val;
28            lazy[2 * i].add_val = lazy[2 * i + 1].add_val = {};
29            lazy[2 * i].has_add = lazy[2 * i + 1].has_add = false;
30            lazy[2 * i].assign_val = lazy[2 * i + 1].assign_val = lazy[i].assign_val;
31            lazy[2 * i].has_assign = lazy[2 * i + 1].has_assign = true;
32        } else {
33            if (lazy[2 * i].has_assign) {
34                tree[2 * i] += s[2 * i] * lazy[i].add_val;
35                lazy[2 * i].assign_val += lazy[i].add_val;
36            } else {
37                tree[2 * i] += s[2 * i] * lazy[i].add_val;
```



```

38     lazy[2 * i].add_val += lazy[i].add_val;
39     lazy[2 * i].has_add = true;
40 }
41 if (lazy[2 * i + 1].has_assign) {
42     tree[2 * i + 1] += s[2 * i + 1] * lazy[i].add_val;
43     lazy[2 * i + 1].assign_val += lazy[i].add_val;
44 } else {
45     tree[2 * i + 1] += s[2 * i + 1] * lazy[i].add_val;
46     lazy[2 * i + 1].add_val += lazy[i].add_val;
47     lazy[2 * i + 1].has_add = true;
48 }
49 }
50 lazy[i].has_assign = lazy[i].has_add = false;
51 lazy[i].assign_val = lazy[i].add_val = {};
52 }
53 void add(int i, int tl, int tr, int l, int r, T addend) {
54     if (l > r) return;
55     if (l == tl && r == tr) {
56         if (lazy[i].has_assign) {
57             tree[i] += s[i] * addend;
58             lazy[i].assign_val += addend;
59         } else {
60             tree[i] += s[i] * addend;
61             lazy[i].add_val += addend;
62             lazy[i].has_add = true;
63         }
64     } else {
65         if (lazy[i].has_update()) push(i);
66         int tm = (tl + tr) / 2;
67         add(2 * i, tl, tm, l, min(r, tm), addend);
68         add(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, addend);
69         tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
70     }
71 }
72 void assign(int i, int tl, int tr, int l, int r, T val) {
73     if (l > r) return;
74     if (l == tl && r == tr) {
75         if (lazy[i].has_add) {
76             lazy[i].has_add = false;
77             lazy[i].add_val = {};
78         }
79         tree[i] = s[i] * val;
80         lazy[i].assign_val = val;
81         lazy[i].has_assign = true;
82     } else {
83         if (lazy[i].has_update()) push(i);
84         int tm = (tl + tr) / 2;
85         assign(2 * i, tl, tm, l, min(r, tm), val);
86         assign(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, val);
87         tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
88     }
89 }
90 T query(int i, int tl, int tr, int l, int r) {
91     if (l > r) return {}; // WARN: depends on `conquer`
92     if (l == tl && r == tr) return tree[i];
93     if (lazy[i].has_update()) push(i);
94     int tm = (tl + tr) / 2;
95     return conquer(
96         query(2 * i, tl, tm, l, min(r, tm)),
97         query(2 * i + 1, tm + 1, tr, max(l, tm + 1), r));
98 }
99
100 public:
101     LazyTree(vector<T> &a)
102     : a(a) {
103         n = ssize(a);
104         s.assign(4 * n, 0);
105         lazy.assign(4 * n, {});
106         tree.assign(4 * n, {});
107         build(1, 0, n - 1);
108     }
109     T conquer(T x, T y) { // WARN: rmb to tweak
110         return x + y;
111     }
112     void add(int l, int r, T addend) { add(1, 0, n - 1, l, r, addend); }
113     void assign(int l, int r, T val) { assign(1, 0, n - 1, l, r, val); }
114     T query(int l, int r) { return query(1, 0, n - 1, l, r); }
115 };

```

11.3.3. Range Addition/Multiplication (Sum Query)

```

1  template <typename T>
2  class LazyTree {
3  public:
4      struct LazyNode {
5          T multiply_val = Mint(1);
6          T add_val;
7          bool has_multiply;
8          bool has_add;
9          bool has_update() { return has_multiply || has_add; }
10     };
11     int n;
12     vector<LazyNode> lazy;
13     vector<Mint> s;
14     vector<T> a, tree;
15     void build(int i, int l, int r) {
16         if (l == r) tree[i] = a[l], s[i] = 1;
17         else {
18             int m = (l + r) / 2;
19             build(2 * i, l, m);
20             build(2 * i + 1, m + 1, r);
21             s[i] = (r - l + 1);
22             tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
23         }
24     }
25     void push(int i) {
26         if (i >= 4 * n) return;
27         if (2 * i < 4 * n) {
28             tree[2 * i] *= lazy[i].multiply_val;
29             tree[2 * i] += lazy[i].add_val;
30             lazy[2 * i].multiply_val *= lazy[i].multiply_val;
31             lazy[2 * i].add_val += lazy[i].add_val;
32             lazy[2 * i].has_multiply = lazy[i].has_multiply;
33             lazy[2 * i].has_add = lazy[i].has_add;
34         }
35         if (2 * i + 1 < 4 * n) {
36             tree[2 * i + 1] *= lazy[i].multiply_val;
37             tree[2 * i + 1] += lazy[i].add_val;
38             lazy[2 * i + 1].multiply_val *= lazy[i].multiply_val;
39             lazy[2 * i + 1].add_val += lazy[i].add_val;
40             lazy[2 * i + 1].has_multiply = lazy[i].has_multiply;
41             lazy[2 * i + 1].has_add = lazy[i].has_add;
42         }
43         lazy[i].has_multiply = lazy[i].has_add = false;
44         lazy[i].multiply_val = Mint(1), lazy[i].add_val = {};
45     }
46     void add(int i, int tl, int tr, int l, int r, T addend) {
47         if (l > r) return;
48         if (l == tl && r == tr) {
49             tree[i] += s[i] * addend;
50             lazy[i].add_val += addend;
51             lazy[i].has_add = true;
52         } else {
53             if (lazy[i].has_update()) push(i);
54             int tm = (tl + tr) / 2;
55             add(2 * i, tl, tm, l, min(r, tm), addend);
56             add(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, addend);
57             tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
58         }
59     }
60     void multiply(int i, int tl, int tr, int l, int r, T val) {
61         if (l > r) return;
62         if (l == tl && r == tr) {
63             tree[i] *= val;
64             lazy[i].add_val *= val, lazy[i].multiply_val *= val;
65             lazy[i].has_multiply = true;
66         } else {
67             if (lazy[i].has_update()) push(i);
68             int tm = (tl + tr) / 2;
69             multiply(2 * i, tl, tm, l, min(r, tm), val);
70             multiply(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, val);
71             tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
72         }
73     }
74     T query(int i, int tl, int tr, int l, int r) {
75         if (l > r) return {}; // WARN: depends on `conquer`
76         if (l == tl && r == tr) return tree[i];
77         if (lazy[i].has_update()) push(i);
78         int tm = (tl + tr) / 2;
79         return conquer(
80             query(2 * i, tl, tm, l, min(r, tm)),
81             query(2 * i + 1, tm + 1, tr, max(l, tm + 1), r));
82     }
83 };

```

```

80     query(2 * i + 1, tm + 1, tr, max(l, tm + 1), r));
81 }
82
83 public:
84     LazyTree(vector<T> &a)
85     : a(a) {
86         n = ssize(a);
87         s.assign(4 * n, 0);
88         lazy.assign(4 * n, {});
89         tree.assign(4 * n, {});
90         build(1, 0, n - 1);
91     }
92     T conquer(T x, T y) { // WARN: rmb to tweak
93         return x + y;
94     }
95     void add(int l, int r, T addend) { add(1, 0, n - 1, l, r, addend); }
96     void multiply(int l, int r, T val) { multiply(1, 0, n - 1, l, r, val); }
97     T query(int l, int r) { return query(1, 0, n - 1, l, r); }
98 };

```

11.3.4. Polynomial Queries

Add [base, base+1, base+2, ...] to the range [l, r].

```

1  class LazyTree {
2  public:
3      vector<ll> s, lazy, d;
4      vector<ll> a, tree;
5      ll sum(ll base, ll diff, ll len) {
6          return (2 * base + diff * (len - 1)) * len / 2;
7      }
8      ll conquer(ll x, ll y) { return x + y; } // WARN: rmb to tweak
9      void build(int i, int l, int r) {
10         if (l == r) tree[i] = a[l], s[i] = 1;
11         else {
12             int m = (l + r) / 2;
13             build(2 * i, l, m);
14             build(2 * i + 1, m + 1, r);
15             s[i] = (r - l + 1);
16             tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
17         }
18     }
19     void push(int i) {
20         tree[2 * i] += sum(lazy[i], d[i], s[2 * i]);
21         tree[2 * i + 1] += sum(s[2 * i] * d[i] + lazy[i], d[i], s[2 * i + 1]);
22         lazy[2 * i] += lazy[i], d[2 * i] += d[i];
23         lazy[2 * i + 1] += s[2 * i] * d[i] + lazy[i], d[2 * i + 1] += d[i];
24         lazy[i] = d[i] = 0;
25     }
26     void
27     add(int i, int tl, int tr, int l, int r,
28         ll base) { // WARN: requires `base` >= 1
29         if (l > r) return;
30         if (l == tl && r == tr) {
31             tree[i] += sum(base, 1, r - l + 1);
32             lazy[i] += base;
33             d[i]++;
34         } else {
35             if (lazy[i]) push(i);
36             int tm = (tl + tr) / 2, gap = max(0, min(r, tm) - l + 1);
37             add(2 * i, tl, tm, l, min(r, tm), base);
38             add(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, base + gap);
39             tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
40         }
41     }
42     ll query(int i, int tl, int tr, int l, int r) {
43         if (l > r) return {}; // WARN: depends on `conquer`
44         if (l == tl && r == tr) return tree[i];
45         if (lazy[i]) push(i);
46         int tm = (tl + tr) / 2;
47         return conquer(
48             query(2 * i, tl, tm, l, min(r, tm)),
49             query(2 * i + 1, tm + 1, tr, max(l, tm + 1), r));
50     }
51
52     public:
53     LazyTree(vector<ll> &a)
54     : a(a) {
55         n = ssize(a);

```

```
56 s.assign(4 * n, 0);
57 lazy.assign(4 * n, {});
58 d.assign(4 * n, {});
59 tree.assign(4 * n, {});
60 build(1, 0, n - 1);
61 }
62 void add(int l, int r, ll base) { add(1, 0, n - 1, l, r, base); }
63 ll query(int l, int r) { return query(1, 0, n - 1, l, r); }
64 };
```

11.3.5. Range Addition (Count Minimums)

query returns the minimum value in a range, as well as the corresponding count.

```
1 class LazyTree {
2 struct LazyNode {
3 int add_val;
4 bool has_add;
5 bool has_update() { return has_add; }
6 };
7 int n;
8 vector<LazyNode> lazy;
9 vector<ii> a, tree;
10 void build(int i, int l, int r) {
11 if (l == r) tree[i] = a[l];
12 else {
13 int m = (l + r) / 2;
14 build(2 * i, l, m);
15 build(2 * i + 1, m + 1, r);
16 tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
17 }
18 }
19 void push(int i) {
20 tree[2 * i][0] += lazy[i].add_val;
21 lazy[2 * i].add_val += lazy[i].add_val;
22 lazy[2 * i].has_add = true;
23 tree[2 * i + 1][0] += lazy[i].add_val;
24 lazy[2 * i + 1].add_val += lazy[i].add_val;
25 lazy[2 * i + 1].has_add = true;
26 lazy[i].has_add = false;
27 lazy[i].add_val = {};
28 }
29 void add(int i, int tl, int tr, int l, int r, int addend) {
30 if (l > r) return;
31 if (l == tl && r == tr) {
32 tree[i][0] += addend;
33 lazy[i].add_val += addend;
34 lazy[i].has_add = true;
35 } else {
36 if (lazy[i].has_update()) push(i);
37 int tm = (tl + tr) / 2;
38 add(2 * i, tl, tm, l, min(r, tm), addend);
39 add(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, addend);
40 tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
41 }
42 }
43 ii query(int i, int tl, int tr, int l, int r) {
44 if (l > r) return { oo, 0 }; // WARN: depends on `conquer`
45 if (l == tl && r == tr) return tree[i];
46 if (lazy[i].has_update()) push(i);
47 int tm = (tl + tr) / 2;
48 return conquer(
49 query(2 * i, tl, tm, l, min(r, tm)),
50 query(2 * i + 1, tm + 1, tr, max(l, tm + 1), r));
51 }
52
53 public:
54 LazyTree(vector<ii> &a)
55 : a(a) {
56 n = ssize(a);
57 lazy.assign(4 * n, {});
58 tree.assign(4 * n, {});
59 build(1, 0, n - 1);
60 }
61 ii conquer(ii x, ii y) { // WARN: rmb to tweak
62 if (x[0] == y[0]) return { x[0], x[1] + y[1] };
63 return min(x, y);
64 }
65 void add(int l, int r, int addend) { add(1, 0, n - 1, l, r, addend); }
```

```
66 ii query(int l, int r) { return query(1, 0, n - 1, l, r); }
67 };
```

11.4. Persistent Segment Tree

Stores earlier versions of the tree. To create a new version, use cpy_root.

```
1 template <typename T>
2 class PersistentTree {
3 struct Node {
4 Node *l, *r;
5 T s;
6 Node(T s)
7 : l(nullptr)
8 , r(nullptr)
9 , s(s) {}
10 Node(Node *u)
11 : l(u->l)
12 , r(u->r)
13 , s(u->s) {}
14 Node(Node *l, Node *r)
15 : l(l)
16 , r(r)
17 , s() { // WARN: depends on `conquer`
18 if (l && r) s = r->s;
19 if (l && !r) s = l->s;
20 if (!l && r) s = conquer(l->s, r->s);
21 }
22 };
23 int n = 0;
24 vector<Node *> roots;
25 Node *build(const vector<T> &a, int tl, int tr) {
26 if (tl == tr) return new Node(a[tl]);
27 int tm = (tl + tr) / 2;
28 return new Node(build(a, tl, tm), build(a, tm + 1, tr));
29 }
30 T query(Node *u, int tl, int tr, int l, int r) {
31 if (l > r) return {}; // WARN: depends on `conquer`
32 if (l == tl && r == tr) return u->s;
33 int tm = (tl + tr) / 2;
34 T ls = query(u->l, tl, tm, l, min(r, tm));
35 T rs = query(u->r, tm + 1, tr, max(l, tm + 1), r);
36 return conquer(ls, rs);
37 }
38 Node *update(Node *u, int tl, int tr, int pos, T s) {
39 if (tl == tr) return new Node(s);
40 int tm = (tl + tr) / 2;
41 if (pos <= tm) return new Node(update(u->l, tl, tm, pos, s), u->r);
42 else return new Node(u->l, update(u->r, tm + 1, tr, pos, s));
43 }
44
45 public:
46 PersistentTree(vector<T> &a, int root_cnt, int init)
47 : roots(root_cnt) {
48 n = ssize(a);
49 roots[init] = build(a, 0, n - 1);
50 }
51 static T conquer(T x, T y) { // WARN: rmb to tweak
52 return x + y;
53 }
54 void update(int root, int pos, T s) {
55 roots[root] = update(roots[root], 0, n - 1, pos, s);
56 }
57 void cpy_root(int src, int dst) { roots[dst] = new Node(roots[src]); }
58 T query(int root, int l, int r) { return query(roots[root], 0, n - 1, l, r); }
59 };
60
61 int n, q, qtype, root_cnt = 0, QMAX = 100'000;
62
63 int main() {
64 cin.tie(nullptr)->sync_with_stdio(false);
65
66 cin >> n;
67 V a(n + 1, 0LL);
68
69 G(i, 1, n) cin >> a[i];
70 PersistentTree ptree(a, QMAX + 1, root_cnt);
71
72 cin >> q;
```

```
73
74 while (q--) {
75 cin >> qtype;
76 if (qtype == 1) {
77 int idx, v;
78 cin >> idx >> v, root_cnt++;
79 ptree.cpy_root(root_cnt - 1, root_cnt);
80 ptree.update(root_cnt, idx, v);
81 } else {
82 int hist_num, l, r;
83 cin >> hist_num >> l >> r;
84 cout << ptree.query(hist_num, l, r) << "\n";
85 }
86 }
87 return 0;
88 };
```

11.5. Merge Sort Tree

Stores the result of merge sort operations; may be used to answer queries such as finding the number of items $\leq x$ within a range.

```
1 struct Data {
2 vector<int> raw;
3 };
4 template <typename T = Data>
5 class MergeSortTree {
6 int n;
7 vector<T> a, tree;
8 void build(int i, int l, int r) {
9 if (l == r) tree[i] = a[l];
10 else {
11 int m = (l + r) / 2;
12 build(2 * i, l, m);
13 build(2 * i + 1, m + 1, r);
14 tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
15 }
16 }
17 void update(int i, int tl, int tr, int pos, T val) {
18 if (tl == tr) tree[i] = val;
19 else {
20 int tm = (tl + tr) / 2;
21 if (pos <= tm) update(2 * i, tl, tm, pos, val);
22 else update(2 * i + 1, tm + 1, tr, pos, val);
23 tree[i] = conquer(tree[2 * i], tree[2 * i + 1]);
24 }
25 }
26 int query(int i, int tl, int tr, int l, int r, int x) {
27 if (l > r) return {}; // WARN: depends on `conquer`
28 if (l == tl && r == tr) {
29 int pos = upper_bound(A(tree[i].raw), x) - begin(tree[i].raw);
30 return pos;
31 }
32 int tm = (tl + tr) / 2;
33 int lans = query(2 * i, tl, tm, l, min(r, tm), x);
34 int rans = query(2 * i + 1, tm + 1, tr, max(l, tm + 1), r, x);
35 return lans + rans;
36 }
37
38 public:
39 MergeSortTree(vector<T> &a)
40 : a(a) {
41 n = ssize(a);
42 tree.assign(4 * n, {});
43 build(1, 0, n - 1);
44 }
45 T conquer(Data x, Data y) { // WARN: rmb to tweak
46 Data ans;
47 merge(A(x.raw), A(y.raw), back_inserter(ans.raw));
48 return ans;
49 }
50 void update(int pos, T val) { update(1, 0, n - 1, pos, val); }
51 int query(int l, int r, int x) { return query(1, 0, n - 1, l, r, x); }
52 };
```

11.6. Li Chao Tree

Stores a collection of lines. Given an x coordinate, returns the minimum y coordinate on any of the lines. In this sense, it maintains the convex hull of the lines.

```

1 using namespace std;
2 typedef long long ll;
3 typedef vector<ll> vll;
4 typedef vector<vll> vvll;
5 typedef pair<ll, ll> pll;
6 #define INF(dtype) numeric_limits<dtype>::max()
7
8 // Create a struct for linear functions
9 // First argument: slope, second argument: y-intercept
10 struct Line {
11     ll m, b;
12
13     ll evaluate(ll x) const {
14         return m * x + b;
15     }
16 };
17
18 /*
19 Maintain the minimum line in the li chao tree
20 */
21 class Tree {
22 public:
23     Tree *lt, *rt;
24     ll l, r;
25     Line v;
26
27     Tree(ll a_l, ll a_r): lt(nullptr), rt(nullptr), l(a_l), r(a_r), v({0ll, INF(ll)}) {}
28
29     void build() {
30         if(l == r) {
31             return;
32         }
33
34         ll m = (l + r) >> 1ll;
35
36         lt = new Tree(l, m);
37         rt = new Tree(m + 1ll, r);
38
39         lt->build();
40         rt->build();
41     }
42
43     void insert_line(Line nl) {
44         ll m = (l + r) >> 1ll;
45         bool less_l = nl.evaluate(l) < v.evaluate(l);
46         bool less_m = nl.evaluate(m + 1ll) < v.evaluate(m + 1ll);
47
48         /*
49          ! TODO why does swap work
50          ! and v = nl does not???
51
52          well, if we do v = nl,
53          we simply erase v from existence
54          we don't want that--we want to consider both
55
56          so nl is considered now
57          therefore we instead propagate v downward
58          */
59         if(less_m) swap(v, nl);
60
61         if(l == r) {
62             return;
63         } else if(less_l != less_m) {
64             lt->insert_line(nl);
65         } else {
66             rt->insert_line(nl);
67         }
68     }
69
70     ll evaluate(ll x) {
71         if(l == r) {
72             return v.evaluate(x);
73         }

```

```

74         ll m = (l + r) >> 1ll;
75
76         if(x <= m) {
77             return min(lt->evaluate(x), v.evaluate(x));
78         } else {
79             return min(rt->evaluate(x), v.evaluate(x));
80         }
81     }
82 };

```

12. Convolution

12.1. Number Theoretic Transform

```

1 constexpr ll MOD = 998'244'353, root = 62;
2 ll modpow(ll b, ll p) {
3     if(p == 0) return 1ll;
4     ll ans = modpow(b, p / 2);
5     ans = ((__int128)(ans)*ans) % MOD;
6     if(p & 1) ans = (ans * b) % MOD;
7     return ans;
8 }
9
10 void ntt(vector<ll> &a) {
11     int n = ssize(a), L = 31 - __builtin_clz(n);
12     static vector<ll> rt(2, 1);
13     for (static int k = 2, s = 2; k < n; k *= 2, s++) {
14         rt.resize(n);
15         ll z[] = { 1, modpow(root, MOD >> s) };
16         for (int i = k; i < 2 * k; i++) rt[i] = rt[i / 2] * z[i & 1] % MOD;
17     }
18     vector<int> rev(n);
19     for (int i = 0; i < n; i++) rev[i] = (rev[i / 2] | (i & 1) << L) / 2;
20     for (int i = 0; i < n; i++)
21         if(i < rev[i]) swap(a[i], a[rev[i]]);
22     for (int k = 1; k < n; k *= 2)
23         for (int i = 0; i < n; i += 2 * k)
24             for (int j = 0; j < k; j++) {
25                 ll z = rt[j + k] * a[i + j + k] % MOD, &ai = a[i + j];
26                 a[i + j + k] = ai - z + (z > ai ? MOD : 0);
27                 ai += (ai + z >= MOD ? z - MOD : z);
28             }
29 }
30
31 vector<ll> conv(const vector<ll> &a, const vector<ll> &b) {
32     if(a.empty() || b.empty()) return {};
33     int s = ssize(a) + ssize(b) - 1, B = 32 - __builtin_clz(s), n = 1 << B;
34     int inv = modpow(n, MOD - 2);
35     vector<ll> left(a), right(b), out(n);
36     left.resize(n), right.resize(n);
37     ntt(left), ntt(right);
38     for (int i = 0; i < n; i++)
39         out[-i & (n - 1)] = (ll)left[i] * right[i] % MOD * inv % MOD;
40     ntt(out);
41     return {out.begin(), out.begin() + s};
42 }

```

13. Math

13.1. Extended Euclidean Algorithm

Finds x and y such that $xa + yb = \gcd(a, b)$.

```

1 class GCD {
2     static pair<ll, ll> solve(ll a, ll b) {
3         if(b == 0) return { 1, 0 };
4         auto [y, x] = solve(b, a % b);
5         y -= a / b * x;
6         return { x, y };
7     }
8
9     public:
10     static pair<ll, ll> extended_gcd(ll a, ll b) {
11         auto [x, y] = solve(a, b);
12         if(x * a + b * y == -gcd(a, b)) x *= -1, y *= -1;
13         return { x, y };

```

```

14     }
15 };

```

13.2. Mint

```

1 class Mint {
2     constexpr ll mod(ll x) const {
3         if(x < 0) x += MOD;
4         return x >= MOD ? x % MOD : x;
5     }
6
7     public:
8     static constexpr ll MOD = 1'000'000'007; // WARN: rmb to tweak
9     ll v;
10     constexpr ll pow(ll p) const {
11         if(!p) return 1;
12         ll ans = pow(p / 2);
13         return mod(p & 1 ? v * mod(ans * ans) : mod(ans * ans));
14     }
15     constexpr ll i() const { return pow(MOD - 2); }
16     constexpr Mint(ll v = 0) : v(mod(v)) {}
17     : v(mod(v)) {}
18     constexpr Mint(const Mint &x) : v(x.v) {}
19     : v(x.v) {}
20     constexpr Mint &operator++() {
21         v = mod(v + 1);
22         return *this;
23     }
24     constexpr Mint operator++(int) {
25         Mint _ = *this;
26         ++(*this);
27         return _;
28     }
29     constexpr Mint &operator--() {
30         v = mod(v - 1);
31         return *this;
32     }
33     constexpr Mint operator--(int) {
34         Mint _ = *this;
35         --(*this);
36         return _;
37     }
38     constexpr Mint &operator=(const Mint &y) {
39         v = y.v;
40         return *this;
41     }
42     constexpr Mint &operator+=(const Mint &y) {
43         v = mod(v + y.v);
44         return *this;
45     }
46     constexpr Mint &operator-=(const Mint &y) {
47         v = mod(v - y.v);
48         return *this;
49     }
50     constexpr Mint &operator*=(const Mint &y) {
51         v = mod(v * y.v);
52         return *this;
53     }
54     constexpr Mint &operator/=(const Mint &y) {
55         v = mod(v * y.i());
56         return *this;
57     }
58     friend constexpr Mint operator+(Mint x, const Mint &y) {
59         x += y;
60         return x;
61     }
62     friend constexpr Mint operator-(Mint x, const Mint &y) {
63         x -= y;
64         return x;
65     }
66     friend constexpr Mint operator*(Mint x, const Mint &y) {
67         x *= y;
68         return x;
69     }
70     friend constexpr Mint operator/(Mint x, const Mint &y) {
71         x /= y;
72         return x;
73     }
74     friend constexpr bool operator==(const Mint &x, const Mint &y) {

```

```
75     return x.v == y.v;
76 }
77 friend constexpr bool operator!=(const Mint &x, const Mint &y) {
78     return x.v != y.v;
79 }
80 friend ostream &operator<<(ostream &os, const Mint &x) { return os << x.v; }
81 ;
```

13.3. Sieve of Eratosthenes

```
1  constexpr int SIEVE_SIZE = 1'000'000; // WARN: rmb to tweak
2  bitset<SIEVE_SIZE + 1> bs;
3  vector<ll> primes;
4  void sieve() {
5      bs.set();
6      bs[0] = bs[1] = false;
7      for (ll i = 2; i <= SIEVE_SIZE; i++)
8          if (bs[i]) {
9              for (ll j = i * i; j <= SIEVE_SIZE; j += i) bs[j] = false;
10             primes.push_back(i);
11         }
12 }
```

13.3.1. Euler's Totient Function

$\varphi(n)$ computes the positive integers up to n that are coprime to n . We can compute it from the prime factors of n .

```
1  ll eulerPhi(ll n) {
2      ll ans = n;
3      for (int i = 0; i < S(primes) && primes[i] * primes[i] <= n; i++) {
4          if (n % primes[i] == 0) ans -= ans / primes[i];
5          while (n % primes[i] == 0) n /= primes[i];
6      }
7      return ans != 1 ? ans - ans / n : ans;
8  }
```

We can also compute it using a modified sieve.

```
1  const int SIEVE_SIZE = 1'000'000; // WARN: rmb to tweak
2  int eulerPhis[SIEVE_SIZE + 1];
3  void sieve() {
4      for (int i = 1; i <= SIEVE_SIZE; i++) eulerPhis[i] = i;
5      for (int i = 2; i <= SIEVE_SIZE; i++)
6          if (eulerPhis[i] == i)
7              for (int j = i; j <= SIEVE_SIZE; j += i) eulerPhis[j] -= eulerPhis[j] / i;
8  }
```

13.4. Fast Factorization

Runs in about $\mathcal{O}(n^{\frac{1}{4}})$ time, to be used sparingly.

```
1  ll modmul(ll x, ll y, ll mod) { return ((__int128)x * y) % mod; }
2
3  ll modpow(ll b, ll p, ll mod) {
4      if (!p) return 1;
5      ll ans = modpow(b, p / 2, mod) % mod;
6      ans = modmul(ans, ans, mod);
7      if (p & 1) ans = modmul(ans, b, mod);
8      return ans;
9  }
10
11 bool is_prime(ll n) {
12     if (n < 2 || n % 6 % 4 != 1) return (n | 1) == 3;
13     ll A[] = { 2, 325, 9375, 28178, 450775, 9780504, 1795265022 },
14         s = __builtin_ctzll(n - 1), d = n >> s;
15     for (ll a : A) {
16         ll p = modpow(a % n, d, n), i = s;
17         while (p != 1 && p != n - 1 && a % n && i-->) p = modmul(p, p, n);
18         if (p != n - 1 && i != s) return 0;
19     }
20     return 1;
21 }
22
23 ll rho(ll n) {
```

```
24     ll x = 0, y = 0, t = 30, prd = 2, i = 1, q;
25     auto f = [&](ll x) { return modmul(x, x, n) + i; };
26     while (t++ % 40 || __gcd(prd, n) == 1) {
27         if (x == y) x = ++i, y = f(x);
28         if ((q = modmul(prd, max(x, y) - min(x, y), n)) prd = q;
29             x = f(x), y = f(f(y));
30     }
31     return __gcd(prd, n);
32 }
33
34 vector<ll> factor(ll n) {
35     if (n == 1) return { 1 };
36     auto solve = [&](auto self, ll n) -> V<ll> {
37         if (n == 1) return {};
38         if (is_prime(n)) return { n };
39         auto d = rho(n);
40         V<ll> a = self(self, d), b = self(self, n / d), ans;
41         for (auto x : a) ans.P(x);
42         for (auto y : b) ans.P(y);
43         return ans;
44     };
45     auto ans = solve(solve, n);
46     return ans;
47 }
```

13.5. Frobenius Number

The *Frobenius number* is the largest integer that *can't* be expressed as a non-negative linear combination of two coprime integers p and q . It is computed as $pq - p - q$.

13.6. Matrix

May be helpful for matrix multiplication problems.

```
1  constexpr ll MOD = 1'000'000'007;
2
3  template <typename T = ll>
4  struct Mat2D {
5      int n, m;
6      vector<vector<T>> > g;
7      Mat2D(int n, int m, bool identity = false)
8          : n(n), m(m),
9            g(n, vector<T>(m)) {
10         if (n != m || !identity) return;
11         for (int i = 0; i < n; i++) {
12             g[i][i] = 1;
13         }
14     }
15
16     constexpr Mat2D pow(ll p) const {
17         if (!p) return Mat2D(this->n, this->m, true);
18         Mat2D ans = pow(p / 2);
19         return p & 1 ? (*this) * ans * ans : ans * ans;
20     }
21
22     constexpr Mat2D &operator*=(const Mat2D &b) {
23         assert(this->m == b.n);
24         Mat2D c = Mat2D(this->n, b.m);
25         for (int i = 0; i < this->n; i++) {
26             for (int j = 0; j < b.m; j++) {
27                 for (int k = 0; k < this->m; k++) {
28                     c.g[i][j] += this->g[i][k] * b.g[k][j];
29                 }
30             }
31         }
32         n = c.n, m = c.m, g = c.g;
33         return *this;
34     }
35
36     friend constexpr Mat2D operator*(Mat2D x, const Mat2D &y) {
37         x *= y;
38         return x;
39     };
```

13.7. Linear Algebra

Useful for dealing with systems of equations. Operations are assumed to be done in modulo p . Formally, operands are elements of $M_{i \times j}(\mathbb{Z}/p\mathbb{Z})$.

```
1  typedef long long ll;
2  typedef vector<ll> vll;
3  typedef vector<vll> vvll;
4
5  // Modular Arithmetic Tools needed for operations in Z/pZ
6  ll fpw(ll base, ll pw, ll md) {
7      base %= md;
8      if (pw == 0) return 1ll;
9      ll bsqr = fpw((base * base) % md, pw >> 1ll, md);
10
11     return (pw & 0b1ll ? (bsqr * base) % md : bsqr);
12 }
13
14 ll minv(ll n, ll md) {
15     return fpw(n, md - 2ll, md);
16 }
17
18 ll frac(ll num, ll den, ll md) {
19     return (num * minv(den, md)) % md;
20 }
21
22 // Prints a matrix
23 void matprint(const vvll& m) {
24     ll r = m.size();
25
26     if (r == 0) {
27         cerr << "empty list!" << endl;
28         return;
29     }
30
31     ll c = m[0ll].size();
32
33     for (ll i = 0; i < r; i++) {
34         for (ll j = 0; j < c; j++) cerr << m[i][j] << " ";
35         cerr << "\n";
36     }
37
38     cerr << flush;
39 }
40
41 // Multiplies two matrices M1 and M2
42 // order of operands follows linear algebra conventions :)
43 vvll matmul(const vvll& m1, const vvll& m2) {
44     ll nr = m1.size(), nc = m2[0].size();
45     ll maxk = m1.size(); // should be equal to m1[0].size()
46
47     vvll res(nr, vll(nc, 0ll));
48
49     for (ll r = 0; r < nr; r++) {
50         for (ll k = 0; k < maxk; k++) {
51             for (ll c = 0; c < nc; c++) {
52                 res[r][c] += m1[r][k] * m2[k][c];
53             }
54         }
55     }
56
57     return res;
58 }
59
60 // Matrix multiplication mod p
61 vvll matmulmod(const vvll& m1, const vvll& m2, ll MOD) {
62     ll nr = m1.size(), nc = m2[0].size();
63     ll maxk = m1.size(); // should be equal to m1[0].size()
64
65     vvll res(nr, vll(nc, 0ll));
66
67
68     for (ll r = 0; r < nr; r++) {
69         for (ll k = 0; k < maxk; k++) {
70             for (ll c = 0; c < nc; c++) {
71                 res[r][c] = (res[r][c] + (m1[r][k] * m2[k][c]) % MOD) % MOD;
72             }
73         }
74     }
75 }
```

```

76     return res;
77 }
78
79 // Matrix-vector multiplication, no modulo
80 vll matvec(const vll& m, const vll& v) {
81     vll res;
82     ll res_size = m.size();
83     ll inp_size = v.size();
84
85     for(ll i = 0; i < res_size; i++) {
86         ll cur_sum = 0ll;
87         for(ll j = 0; j < inp_size; j++) {
88             cur_sum += v[j] * m[i][j];
89         }
90         res.push_back(cur_sum);
91     }
92
93     return res;
94 }
95
96 // Matrix-vector multiplication, with modulo
97 vll matvecmod(const vll& m, const vll& v, ll MOD) {
98     vll res;
99     ll res_size = m.size();
100    ll inp_size = v.size();
101
102    for(ll i = 0; i < res_size; i++) {
103        ll cur_sum = 0ll;
104        for(ll j = 0; j < inp_size; j++) {
105            cur_sum = (cur_sum + (v[j] * m[i][j]) % MOD) % MOD;
106        }
107        res.push_back(cur_sum);
108    }
109
110    return res;
111 }
112
113 // Creates an identity matrix of size n x n
114 vll id_mat(ll n) {
115     vll res;
116     for(ll i = 0; i < n; i++) {
117         vll res_row(n, 0ll);
118         res.push_back(res_row);
119     }
120     for(ll i = 0; i < n; i++) {
121         res[i][i] = 1ll;
122     }
123     return res;
124 }
125
126 // Fast matrix exponentiation
127 // m has to be a square matrix
128 // O(n^3 log pw)
129 vll matpowmod(const vll& m, ll pw, ll md) {
130     if(pw == 0) {
131         return id_mat(m.size());
132     }
133     vll sqr_sqrt = matpowmod(matmulmod(m, m, md), pw >> 1, md);
134     return (pw & 1 ? matmulmod(sqr_sqrt, m, md) : sqr_sqrt);
135 }
136
137 // Matrix Addition
138 vll matadd(const vll& m1, const vll& m2, ll md) {
139     ll r1 = m1.size(), c1 = m1[0ll].size();
140     ll r2 = m2.size(), c2 = m2[0ll].size();
141
142     assert(r1 == r2 && c1 == c2);
143
144     vll res(r1, vll(c1, 0ll));
145
146     for(ll i = 0ll; i < r1; i++) for(ll j = 0ll; j < c1; j++) res[i][j] = (m1[i][j] + m2[i][j]) % md;
147
148     return res;
149 }
150
151 // Matrix Subtraction
152 vll matsub(const vll& m1, const vll& m2, ll md) {
153     ll r1 = m1.size(), c1 = m1[0ll].size();
154     ll r2 = m2.size(), c2 = m2[0ll].size();
155

```

```

156     assert(r1 == r2 && c1 == c2);
157
158     vll res(r1, vll(c1, 0ll));
159
160     for(ll i = 0ll; i < r1; i++) for(ll j = 0ll; j < c1; j++) res[i][j] = ((m1[i][j] - m2[i][j]) % md + md) % md;
161
162     return res;
163 }
164
165 // Copy a matrix
166 vll matcopy(const vll& mat) {
167     ll r = mat.size(), c = mat[0ll].size();
168
169     vll res(r, vll(c, 0ll));
170
171     for(ll i = 0ll; i < r; i++) for(ll j = 0ll; j < c; j++) res[i][j] = mat[i][j];
172
173     return res;
174 }
175
176 // Matrix Transpose
177 vll transpose(const vll& mat) {
178     ll r = mat.size();
179
180     vll empty;
181     if(r == 0ll) return empty;
182
183     ll c = mat[0ll].size();
184
185     if(c == 0ll) return empty;
186
187     vll res(c, vll(r, 0ll));
188     for(ll i = 0ll; i < r; i++) for(ll j = 0ll; j < c; j++) res[j][i] = mat[i][j];
189
190     return res;
191 }
192
193 // In-place Gauss-Jordan Elimination mod p
194 // lc is the number of columns to reduce. useful for reducing augmented matrices
195 // returns the determinant
196 ll gj_inplace(vll& mat, ll lc, ll md) {
197     ll r = mat.size();
198
199     if(r == 0ll) return 0ll;
200     ll c = mat[0ll].size();
201     if(c == 0ll) return 0ll;
202
203     ll det = 1ll;
204
205     // reducing to row-echelon form
206     ll cc = 0ll;
207     for(ll cr = 0ll; cr < r && cc < min(c, lc); cr++) {
208         // find the first row in [i, r] that's nonzero
209         ll fstnzero = -1ll;
210
211         do {
212             for(ll j = cr; j < r; j++) if(mat[j][cc] != 0ll) {fstnzero = j; break;}
213
214             if(fstnzero == -1ll) {
215                 cc++;
216                 det = 0ll;
217             }
218             while(fstnzero == -1ll && cc < c);
219
220             if(cc >= c) break;
221
222             if(cr != fstnzero) {
223                 swap(mat[cr], mat[fstnzero]);
224                 det = (md - det);
225             }
226
227             // normalize the current row
228             const ll lead_inv = minv(mat[cr][cc], md);
229
230             det = (det * mat[cr][cc]) % md;
231             for(ll j = cc; j < c; j++) {
232                 mat[cr][j] = (mat[cr][j] * lead_inv) % md;
233
234                 mat[cr][j] %= md;
235                 mat[cr][j] += md;

```

```

236         mat[cr][j] %= md;
237     }
238
239     // kill all terms below and above
240
241     for(ll j = 0ll; j < r; j++) {
242         if(j == cr) continue;
243         ll cur_lead_entry = mat[j][cc];
244         if(cur_lead_entry == 0ll) continue;
245
246         ll multi = (cur_lead_entry) % md;
247
248         for(ll i = cc; i < c; i++) {
249             mat[j][i] -= (mat[cr][i] * multi) % md;
250             mat[j][i] %= md;
251             mat[j][i] += md;
252             mat[j][i] %= md;
253         }
254     }
255
256     cc++;
257 }
258
259 return det;
260 }
261
262 // Gauss-Jordan Elimination mod p
263 vll gj(const vll& mat, ll lc, ll md) {
264     vll res = matcopy(mat);
265
266     gj_inplace(res, lc, md);
267
268     return res;
269 }
270
271 // Determinant Computation (Using Gauss-Jordan Elimination)
272 ll determinant(const vll& mat, ll md) {
273     vll matcpy = matcopy(mat);
274
275     return gj_inplace(matcpy, matcpy[0ll].size(), md);
276 }
277
278 // Row Space Basis Finder (i.e., nonzero row selector)
279 vll rowspace(const vll& mat, ll md) {
280     vll matcpy = matcopy(mat);
281
282     ll r = matcpy.size(), c = matcpy[0ll].size();
283     gj_inplace(matcpy, c, md);
284
285     vll res;
286
287     for(ll i = 0ll; i < r; i++) {
288         bool all_zero = true;
289
290         for(ll j = 0ll; j < c; j++) if(matcpy[i][j] != 0ll) {all_zero = false; break;}
291
292         if(!all_zero) res.push_back(matcpy[i]);
293     }
294
295     return res;
296 }
297
298 // Matrix Rank Finder (Note: rank(A) = dim(Row(A)) = dim(Col(A)))
299 ll matrank(const vll& mat, ll md) {
300     if((ll)(mat.size()) == 0ll) return 0ll;
301     if((ll)(mat[0ll].size()) == 0ll) return 0ll;
302     vll matcpy = matcopy(mat);
303
304     ll r = matcpy.size(), c = matcpy[0ll].size();
305     gj_inplace(matcpy, c, md);
306
307     ll res = 0ll;
308
309     for(ll i = 0ll; i < r; i++) {
310         bool all_zero = true;
311
312         for(ll j = 0ll; j < c; j++) if(matcpy[i][j] != 0ll) {all_zero = false; break;}
313

```



```

316     if(!all_zero) res++;
317 }
318
319 return res;
320 }
321
322 // Col Space Basis Finder (i.e., leading entry column selector)
323 // Note: Returns the column space as row vectors!
324 vll colspace(const vll& mat, ll md) {
325     vll matcpy = matcopy(mat);
326
327     ll r = matcpy.size(), c = matcpy[0ll].size();
328
329     // reducing to row-echelon form
330     vll res;
331
332     ll cc = 0ll;
333     for(ll cr = 0ll; cr < r && cc < c; cr++) {
334         // find the first row in [i, r] that's nonzero
335         ll fstnonzero = -1ll;
336
337         do {
338             for(ll j = cr; j < r; j++) if(matcpy[j][cc] != 0ll) {fstnonzero = j; break;}
339
340             if(fstnonzero == -1ll) {
341                 cc++;
342             } while(fstnonzero == -1ll && cc < c);
343
344             if(cc >= c) break;
345
346             if(cr != fstnonzero) {
347                 swap(matcpy[cr], matcpy[fstnonzero]);
348             }
349
350             // normalize the current row
351             const ll lead_inv = minv(matcpy[cr][cc], md);
352
353             for(ll j = cc; j < c; j++) {
354                 matcpy[cr][j] = (matcpy[cr][j] * lead_inv) % md;
355
356                 matcpy[cr][j] %= md;
357                 matcpy[cr][j] += md;
358                 matcpy[cr][j] %= md;
359             }
360
361             // kill all terms below and above
362
363             for(ll j = 0ll; j < r; j++) {
364                 if(j == cr) continue;
365                 ll cur_lead_entry = matcpy[j][cc];
366                 if(cur_lead_entry == 0ll) continue;
367
368                 ll multi = (cur_lead_entry) % md;
369
370                 for(ll i = cc; i < c; i++) {
371                     matcpy[j][i] -= (matcpy[cr][i] * multi) % md;
372                     matcpy[j][i] %= md;
373                     matcpy[j][i] += md;
374                     matcpy[j][i] %= md;
375                 }
376             }
377
378             // Adding the current column with a leading entry to the column basis
379             vll next_vec(r, 0ll);
380
381             for(ll i = 0ll; i < r; i++) next_vec[i] = mat[i][cc];
382
383             res.push_back(next_vec);
384
385             cc++;
386         }
387
388         return res;
389     }
390 }
391
392 // Null Space Basis Finder
393 vll nullspace(const vll& mat, ll md) {
394     vll matcpy = matcopy(mat);

```

```

396     gj_inplace(matcpy, matcpy[0ll].size(), md);
397
398     // Find the indices of the free parameters
399     ll r = matcpy.size(), c = matcpy[0ll].size();
400     vll leads;
401     vll freeparams;
402     set<ll> freeparamsset;
403
404     ll cur_xpos = 0ll;
405     for(ll i = 0ll; i < r; i++) {
406         while(cur_xpos < c && matcpy[i][cur_xpos] == 0ll) {
407             cur_xpos++;
408         }
409
410         if(cur_xpos == c) break;
411
412         leads.push_back(cur_xpos);
413     }
414
415     ll cur_ind = 0ll;
416
417     ll num_leads = leads.size();
418
419     for(ll i = 0ll; i < c; i++) {
420         if(cur_ind < num_leads && i == leads[cur_ind]) {
421             cur_ind++;
422         } else {
423             freeparams.push_back(i);
424             freeparamsset.insert(i);
425         }
426     }
427
428     // For each free parameter, put a default value into the corresponding vector
429     ll num_free_params = freeparams.size();
430
431     if(num_free_params == 0ll) {
432         vll empty;
433         return empty;
434     }
435
436     vll res(num_free_params, vll(c, 0ll));
437
438     for(ll i = 0ll; i < num_free_params; i++) {
439         res[i][freeparams[i]] = 1ll;
440     }
441
442     // Process each leading entry
443     cur_xpos = 0ll;
444     for(ll i = 0ll; i < r; i++) {
445         while(cur_xpos < c && matcpy[i][cur_xpos] == 0ll) {
446             cur_xpos++;
447         }
448
449         if(cur_xpos == c) break;
450
451         ll cur_free_param_ind = 0ll;
452         for(ll j = 0ll; j < c && cur_free_param_ind < num_free_params; j++) {
453             if((ll)(freeparamsset.count(j)) > 0ll) {
454                 if(j > cur_xpos) {
455                     res[cur_free_param_ind][cur_xpos] = (md - matcpy[i][j]) % md;
456                 }
457
458                 cur_free_param_ind++;
459             }
460         }
461     }
462
463     return res;
464 }
465
466 // Matrix Inverter
467 vll matinv(const vll& mat, ll md) {
468     ll r = mat.size(), c = mat[0ll].size();
469     vll augmented(r, vll(c << 1ll, 0ll));
470
471     for(ll i = 0ll; i < r; i++) for(ll j = 0ll; j < c; j++) augmented[i][j] = mat[i][j];
472
473     for(ll i = 0ll; i < r; i++) augmented[i][i + c] = 1ll;
474
475     gj_inplace(augmented, c, md);

```

```

476     vll res(r, vll(c, 0ll));
477
478     for(ll i = 0ll; i < r; i++) for(ll j = 0ll; j < c; j++) res[i][j] = augmented[i][j + c];
479
480     return res;
481 }
482
483 // Eigenvector Finder given Eigenvalue
484 // (Could be used to find the stationary distribution of a markov chain)
485 vll find_eigenvectors(const vll& mat, ll lambda, ll md) {
486     vll matcpy = matcopy(mat);
487
488     ll r = matcpy.size();
489     assert(r > 0ll);
490     ll c = matcpy[0ll].size();
491     assert(c > 0ll);
492
493     for(ll i = 0ll; i < r; i++) {
494         matcpy[i][i] -= lambda;
495         matcpy[i][i] %= md;
496         matcpy[i][i] += md;
497         matcpy[i][i] %= md;
498     }
499
500     vll eigenvecs = nullspace(matcpy, md);
501
502     return eigenvecs;
503 }
504
505 // Find stationary distribution of a markov chain
506 vll find_stationary_distrib(const vll& tr_mat, ll md) {
507     vll vecs = find_eigenvectors(transpose(tr_mat), 1ll, md);
508
509     ll numvecs = vecs.size();
510
511     if(numvecs == 0ll) {
512         vll empty;
513         return empty;
514     }
515
516     ll vecdim = vecs[0ll].size();
517
518     for(ll i = 0ll; i < numvecs; i++) {
519         ll sum_entries = 0ll;
520
521         for(ll j = 0ll; j < vecdim; j++) {
522             sum_entries += vecs[i][j];
523             sum_entries %= md;
524             sum_entries += md;
525             sum_entries %= md;
526         }
527
528         for(ll j = 0ll; j < vecdim; j++) {
529             vecs[i][j] *= minv(sum_entries, md);
530             vecs[i][j] %= md;
531             vecs[i][j] += md;
532             vecs[i][j] %= md;
533         }
534
535         return vecs;
536     }
537
538     // Check basis b1 is subset of b2
539     bool is_basis_subset(const vll& b1, const vll& b2, ll md) {
540         ll lc = b2.size();
541         ll b1s = b1.size();
542
543         vll concat = b2;
544         for(ll i = 0ll; i < b1s; i++) concat.push_back(b1[i]);
545
546         vll reducedmat = gj(transpose(concat), lc, md);
547
548         ll nr = reducedmat.size();
549
550         for(ll i = 0ll; i < nr; i++) {
551             bool all_left_zeros = true;
552             bool all_right_zeros = true;

```

```

556     for(ll j = 0ll; j < lc; j++) {
557         if(reducedmat[i][j] != 0ll) {
558             all_left_zeros = false;
559             break;
560         }
561     }
562     for(ll j = lc; j < lc + b1s; j++) {
563         if(reducedmat[i][j] != 0ll) {
564             all_right_zeros = false;
565             break;
566         }
567     }
568 }
569
570 // if left does not imply right, return false (formal negation of implication)
571 if(all_left_zeros && !all_right_zeros) return false;
572 }
573
574 return true;
575 }
576
577 // Check if Two Bases are Equal
578 bool is_basis_eq(const vvl& b1, const vvl& b2, ll md) {
579     return is_basis_subset(b1, b2, md) && is_basis_subset(b2, b1, md);
580 }
581
582 // Check if Vector in Span (equivalent to checking if the span of the singular vector is a subset of
the set of vectors)
583 bool is_in_span(const vll& v, const vvl& b, ll md) {
584     vvl b1;
585     b1.push_back(v);
586
587     return is_basis_subset(b1, b, md);
588 }

```

14. Strings

14.1. String Splitting

Because C++ doesn't offer a split function by default, we may have to write our own sometimes.

```

1 auto string_split = [](const string &s, const string &delims) {
2     size_t prv = 0, pos;
3     vector<string> parsed;
4     while ((pos = s.find_first_of(delims, prv)) != string::npos) {
5         if (pos > prv) parsed.push_back(s.substr(prv, pos - prv));
6         prv = pos + 1;
7     }
8     if (prv < size(s)) parsed.push_back(s.substr(prv, string::npos));
9     return parsed;
10 };

```

14.2. Hashing

You may use two different prime numbers to create a stronger hash.

```

1 int NMAX = 500'000; // WARN: rmb to tweak
2 vector<Mint> pwr(NMAX), invs(NMAX);
3
4 class Shash {
5     vector<Mint> p;
6
7 public:
8     static constexpr Mint pwr = 31;
9     static constexpr Mint inv = pwr.i();
10
11     Shash(string s) {
12         p.resize(S(s));
13         for (int i = 0; i < S(s); i++) {
14             p[i] = pwr[s[i] * (s[i] - 'a' + 1)];
15         }
16         partial_sum(begin(p), end(p), begin(p));
17
18     Mint query(int l, int r) {
19         if (l == 0) return p[r];
20         return (p[r] - p[l - 1]) * invs[l];

```

```

20     }
21 };
22
23 int main() {
24     Mint pw = 1, in = 1;
25     for (int i = 0; i < NMAX; i++) {
26         pwr[i] = pw, invs[i] = in;
27         pw *= Shash::pwr, in *= Shash::inv;
28     }
29 }

```

14.3. Manacher's Algorithm

Computes the maximum length of a palindrome centered at any index.

```

1 class Manacher {
2     deque<int> p;
3     deque<int> manacher_odd(string s) {
4         int n = ssize(s);
5         s = '$' + s + '$';
6         deque<int> p(n + 2);
7         int l = 1, r = 1;
8         for (int i = 1; i <= n; i++) {
9             p[i] = max(0, min(r - i, p[l + (r - i)]));
10            while (s[i - p[i]] == s[i + p[i]]) p[i]++;
11            if (i + p[i] > r) l = i - p[i], r = i + p[i];
12        }
13        p.pop_front(), p.pop_back();
14        return p;
15    }
16
17 public:
18     Manacher(const string &s) {
19         string t(1, '|');
20         for (auto &c : s) t += c, t += '|';
21         auto ans = manacher_odd(t);
22         ans.pop_front(), ans.pop_back();
23         p = ans;
24     }
25     int get_len(int i, bool is_odd) {
26         // NOTE: center palindrome at `i` and `i+1` if `is_odd` is false
27         return is_odd ? 2 * (p[2 * i] / 2 - 1) + 1 : p[2 * i + 1] - 1;
28     }
29 };

```

14.4. Z-function

The *Z-function* of a string is an array of length n where the i -th element is equal to the greatest number of characters starting from position i that coincides with the first characters of the string.

```

1 auto z_function = [](const string &s) -> vector<int> {
2     int l = 0, r = 0, n = ssize(s);
3     vector<int> z(n);
4     for (int i = 1; i < n; i++) {
5         if (i < r) z[i] = min(r - i, z[i - l]);
6         while (i + z[i] < n && s[z[i]] == s[i + z[i]]) z[i]++;
7         if (i + z[i] > r) l = i, r = i + z[i];
8     }
9     return z;
10 };

```

14.5. Trie

```

1 constexpr int K = 26;
2
3 struct Vertex {
4     vector<int> nxt;
5     bool is_end;
6     int cnt;
7     Vertex()
8         : nxt(K, -1)
9         , is_end(false)
10        , cnt(0) {}
11 };
12

```

```

13 vector<Vertex> trie(1);
14
15 void insert(const string &s) {
16     int u = 0;
17     for (const auto &c : s) {
18         int i = c - 'a';
19         if (trie[u].nxt[i] == -1) {
20             trie[u].nxt[i] = ssize(trie);
21             trie.emplace_back();
22         }
23         u = trie[u].nxt[i];
24         trie[u].cnt++;
25     }
26     trie[u].is_end = true;
27 }
28
29 ll search(const string &s) {
30     int u = 0;
31     ll ans = 0;
32     for (const auto &c : s) {
33         int i = c - 'a';
34         if (trie[u].nxt[i] == -1) break;
35         u = trie[u].nxt[i];
36         ans += trie[u].cnt;
37     }
38     return ans;
39 }

```

You can also find the maximum XOR of a subarray.

```

1 class Trie {
2     int max_length;
3     int node_cnt;
4     vector<int> stop;
5     vector<vector<int>> trie;
6
7 public:
8     Trie(int max_length)
9         : max_length(max_length)
10        , node_cnt(0)
11        , stop(max_length + 1, false)
12        , trie(max_length, vector<int>(2)) {}
13     void insert(int word) {
14         int node = 0;
15         for (int i = 30; i >= 0; i--) {
16             int b = (word >> i) & 1;
17             if (!trie[node][b]) trie[node][b] = ++node_cnt;
18             node = trie[node][b];
19         }
20         stop[node] = true;
21     }
22     int find(int word) {
23         int node = 0, s = 0;
24         for (int i = 30; i >= 0; i--) {
25             int b = (word >> i) & 1;
26             if (trie[node][b ^ 1]) {
27                 if (b ^ 1) s |= (1 << i);
28                 node = trie[node][b ^ 1];
29             } else {
30                 if (b) s |= (1 << i);
31                 node = trie[node][b];
32             }
33         }
34         return s;
35     }
36 };

```

14.6. Suffix Array

A *suffix array* contains integers that represent the *starting indexes* of all suffixes of a given string, in sorted order.

```

1 class SuffixArray {
2     static constexpr int alphabet = 256;
3     static vector<int> sort_cyclic_shifts(const string &s) {
4         int n = ssize(s), classes = 1;
5         vector<int> p(n), c(n), pn(n), cn(n), cnt(max(alphabet, n), 0);

```

```

6   for (int i = 0; i < n; i++) cnt[s[i]]++;
7   for (int i = 1; i < alphabet; i++) cnt[i] += cnt[i - 1];
8   for (int i = 0; i < n; i++) p[-cnt[s[i]]] = i;
9   c[p[0]] = 0;
10  for (int i = 1; i < n; i++) {
11      if (s[p[i]] != s[p[i - 1]]) classes++;
12      c[p[i]] = classes - 1;
13  }
14  for (int h = 0; (1 << h) < n; h++) {
15      for (int i = 0; i < n; i++) {
16          pn[i] = p[i] - (1 << h);
17          if (pn[i] < 0) pn[i] += n;
18      }
19      fill(begin(cnt), begin(cnt) + classes, 0);
20      for (int i = 0; i < n; i++) cnt[c[pn[i]]]++;
21      for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
22      for (int i = n - 1; i >= 0; i--) p[-cnt[c[pn[i]]]] = pn[i];
23      cn[p[0]] = 0;
24      classes = 1;
25      for (int i = 1; i < n; i++) {
26          pair<int, int> cur = { c[p[i]], c[(p[i] + (1 << h)) % n] };
27          pair<int, int> prv = { c[p[i - 1]], c[(p[i - 1] + (1 << h)) % n] };
28          if (cur != prv) classes++;
29          cn[p[i]] = classes - 1;
30      }
31      c.swap(cn);
32  }
33  return p;
34 }
35
36 public:
37 static vector<int> build(const string &_) {
38     string s = _ + '$';
39     vector<int> a = sort_cyclic_shifts(s);
40     a.erase(begin(a));
41     return a;
42 }
43 };
44
45 int n;
46 string s;
47 auto sar = SuffixArray::build(s);
48
49 auto compare = [](const string &t, int suf_idx) -> int {
50     for (int i = 0; i < min((int)s.size(t), n - suf_idx); i++) {
51         if (t[i] < s[suf_idx + i]) return -1;
52         if (t[i] > s[suf_idx + i]) return 1;
53         if (i == s.size(t) - 1) return 0;
54     }
55     return 1;
56 };
57
58 auto count_substr = [](const string &t) -> int {
59     // WARN: 'sar' := suffix array (declared in the outer scope)
60     int lo = 0, hi = n - 1, mi, an = -1, na = -1;
61     while (lo <= hi) {
62         mi = midpoint(lo, hi);
63         if (compare(t, sar[mi]) > 0) an = mi, lo = mi + 1;
64         else hi = mi - 1;
65     }
66     lo = 0, hi = n - 1;
67     while (lo <= hi) {
68         mi = midpoint(lo, hi);
69         if (compare(t, sar[mi]) >= 0) na = mi, lo = mi + 1;
70         else hi = mi - 1;
71     }
72     return na - an;
73 };

```

You can also modify it to retrieve the length of the longest common prefix between two suffixes starting at i and j .

```

1  class SuffixArray {
2      static constexpr int alphabet = 256;
3
4  public:
5      static int n;
6      static vector<vector<int>> cmat;
7      static vector<int> sort_cyclic_shifts(const string &s) {

```

```

8      n = s.size();
9      int classes = 1;
10     vector<int> p(n), c(n), pn(n), cn(n), cnt(max(alphabet, n), 0);
11     for (int i = 0; i < n; i++) cnt[s[i]]++;
12     for (int i = 1; i < alphabet; i++) cnt[i] += cnt[i - 1];
13     for (int i = 0; i < n; i++) p[-cnt[s[i]]] = i;
14     c[p[0]] = 0;
15     for (int i = 1; i < n; i++) {
16         if (s[p[i]] != s[p[i - 1]]) classes++;
17         c[p[i]] = classes - 1;
18     }
19     cmat.P(c);
20     for (int h = 0; (1 << h) < n; h++) {
21         for (int i = 0; i < n; i++) {
22             pn[i] = p[i] - (1 << h);
23             if (pn[i] < 0) pn[i] += n;
24         }
25         fill(begin(cnt), begin(cnt) + classes, 0);
26         for (int i = 0; i < n; i++) cnt[c[pn[i]]]++;
27         for (int i = 1; i < classes; i++) cnt[i] += cnt[i - 1];
28         for (int i = n - 1; i >= 0; i--) p[-cnt[c[pn[i]]]] = pn[i];
29         cn[p[0]] = 0;
30         classes = 1;
31         for (int i = 1; i < n; i++) {
32             pair<int, int> cur = { c[p[i]], c[(p[i] + (1 << h)) % n] };
33             pair<int, int> prv = { c[p[i - 1]], c[(p[i - 1] + (1 << h)) % n] };
34             if (cur != prv) classes++;
35             cn[p[i]] = classes - 1;
36         }
37         c.swap(cn);
38         cmat.P(c);
39     }
40     return p;
41 }
42 static int get_lcp(int i, int j) { // WARN: doesn't work for 'i == j'
43     int ans = 0;
44     for (int k = __lg(n); k >= 0; k--) {
45         if (cmat[k][i % n] == cmat[k][j % n]) {
46             ans += (1 << k);
47             i += (1 << k);
48             j += (1 << k);
49         }
50     }
51     return ans;
52 }
53 static vector<int> build(const string &_) { // NOTE: call at the start
54     string s = _ + '$';
55     vector<int> a = sort_cyclic_shifts(s);
56     a.erase(begin(a));
57     return a;
58 }
59 };
60 int SuffixArray::n;
61 vector<vector<int>> SuffixArray::cmat;

```

14.7. Aho-Corasick Algorithm

This variant checks if any substring of a string s matches any string in the dictionary.

```

1  const int NMAX = 1'000 * 100; // WARN: rmb to tweak
2
3  int trie[NMAX + 1][26], fail[NMAX + 1], stop[NMAX + 1], idx;
4
5  void insert(const string &s) {
6      int i = 0;
7      for (auto &c : s) {
8          if (!trie[i][c - 'a']) trie[i][c - 'a'] = ++idx;
9          i = trie[i][c - 'a'];
10     }
11     stop[i] = true;
12 }
13 void build() { // NOTE: construct suffix link(s)
14     queue<int> q;
15     for (int i = 0; i < 26; i++)
16         if (trie[0][i]) q.push(trie[0][i]);
17     while (!empty(q)) {
18         int u = q.front();
19         q.pop();

```

```

20     stop[u] |= stop[fail[u]];
21     for (int i = 0; i < 26; i++) {
22         if (trie[u][i]) {
23             fail[trie[u][i]] = trie[fail[u]][i];
24             q.push(trie[u][i]);
25         } else {
26             trie[u][i] = trie[fail[u]][i];
27         }
28     }
29 }
30 }
31 bool search(const string &s) {
32     int i = 0;
33     for (auto &c : s) {
34         i = trie[i][c - 'a'];
35         if (stop[i]) return true;
36     }
37     return false;
38 }

```